Laser based Directed Energy Deposition

Laser based Directed Energy Deposition

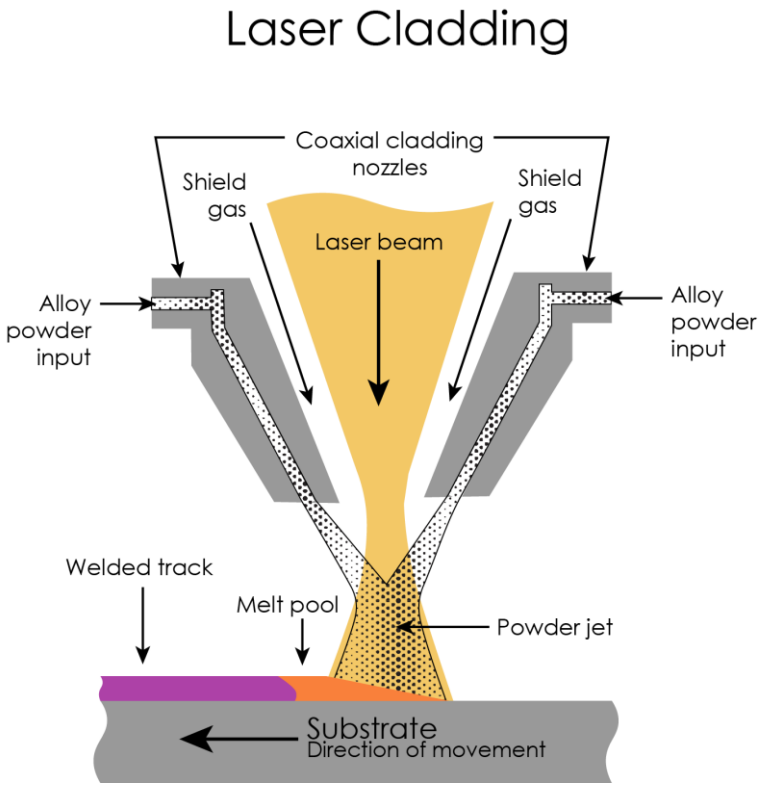
Synonyms in Literature

- ✓ Laser Metal Deposition (LMD)
- ✓ Laser Engineered Net Shaping (LENS)
- ✓ Direct Metal Deposition (DMD)
- ✓ Direct Laser Deposition (DLD)
- ✓ Laser Deposition Welding (LDW)
- ✓ Directed Light Fabrication (DLF)
- ✓ Laser Assisted Metal Deposition

Extended version of Laser Cladding : 3D Laser Cladding

Laser based Directed Energy Deposition

Laser Cladding



- ✓ Surface modification process
- ✓ Laser based material deposition / welding procedure
- ✓ Tribological coating / Corrosion protection

Video Demonstration

Laser based Directed Energy Deposition

Directed Energy Deposition is defined as *“an additive manufacturing process in which focused thermal energy is used to fuse materials by melting as they are being deposited”.*

Process Conditions

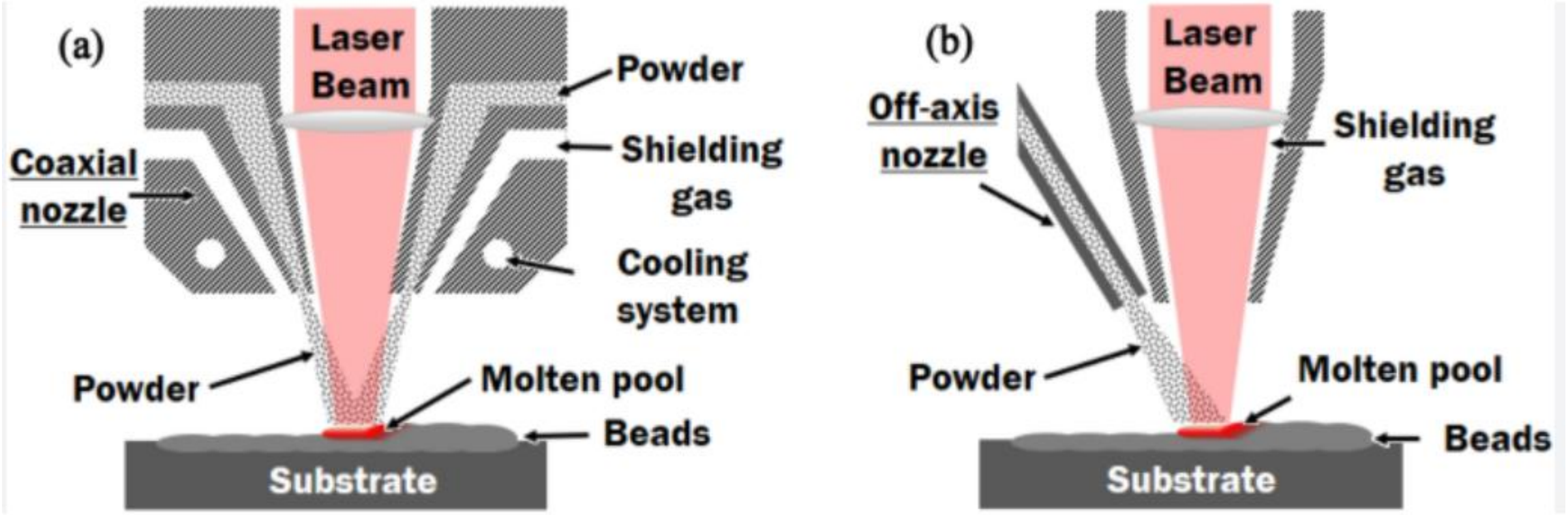
- ✓ Focused delivery of thermal energy (Directed Energy)
- ✓ Fusion by complete melting
- ✓ Fusion during the deposition

Directed energy deposition process can be

- ✓ Powder fed deposition (PFD)
- ✓ Wire fed deposition (WFD)

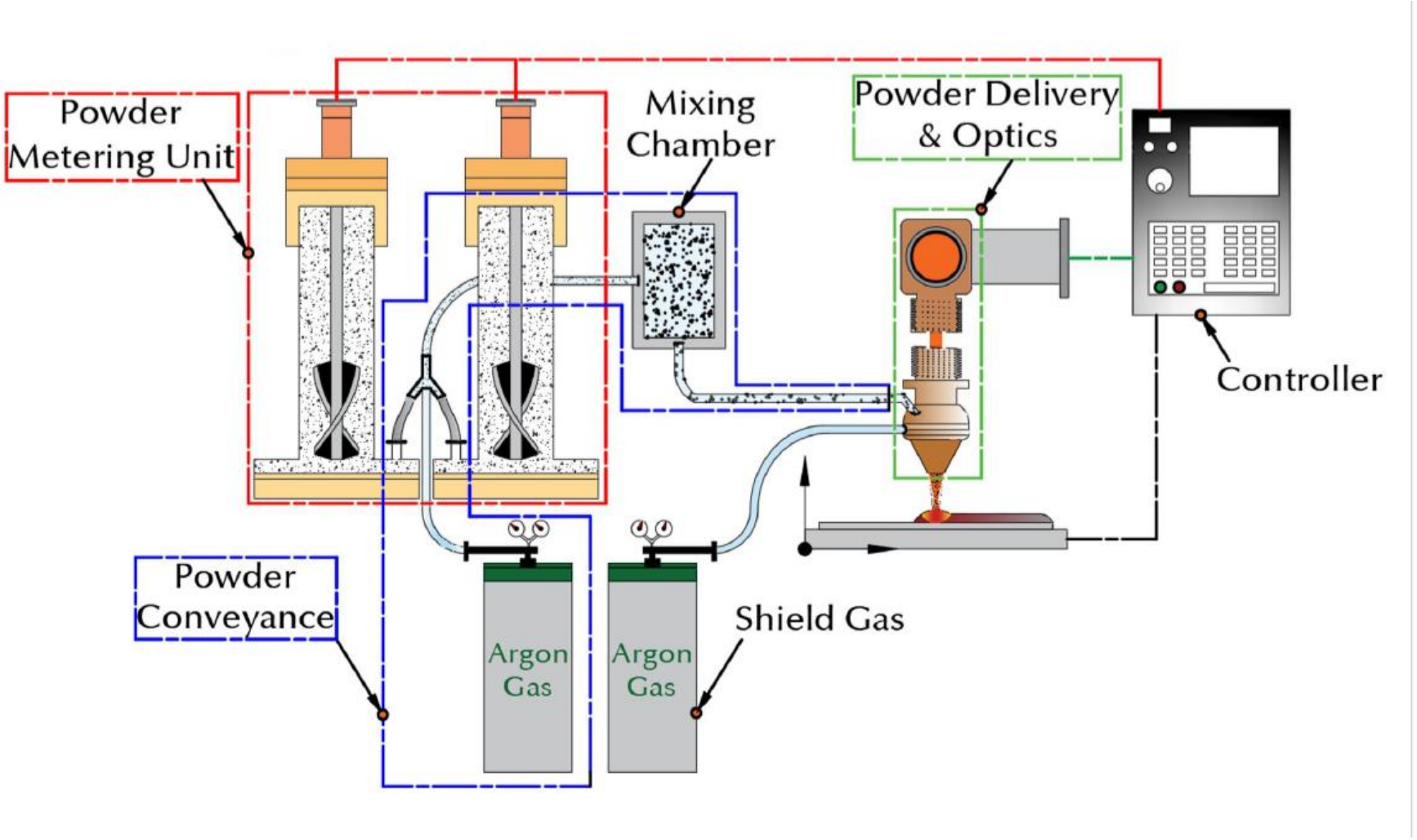
Laser based Directed Energy Deposition

Powder Feed Deposition : Nozzle Configurations



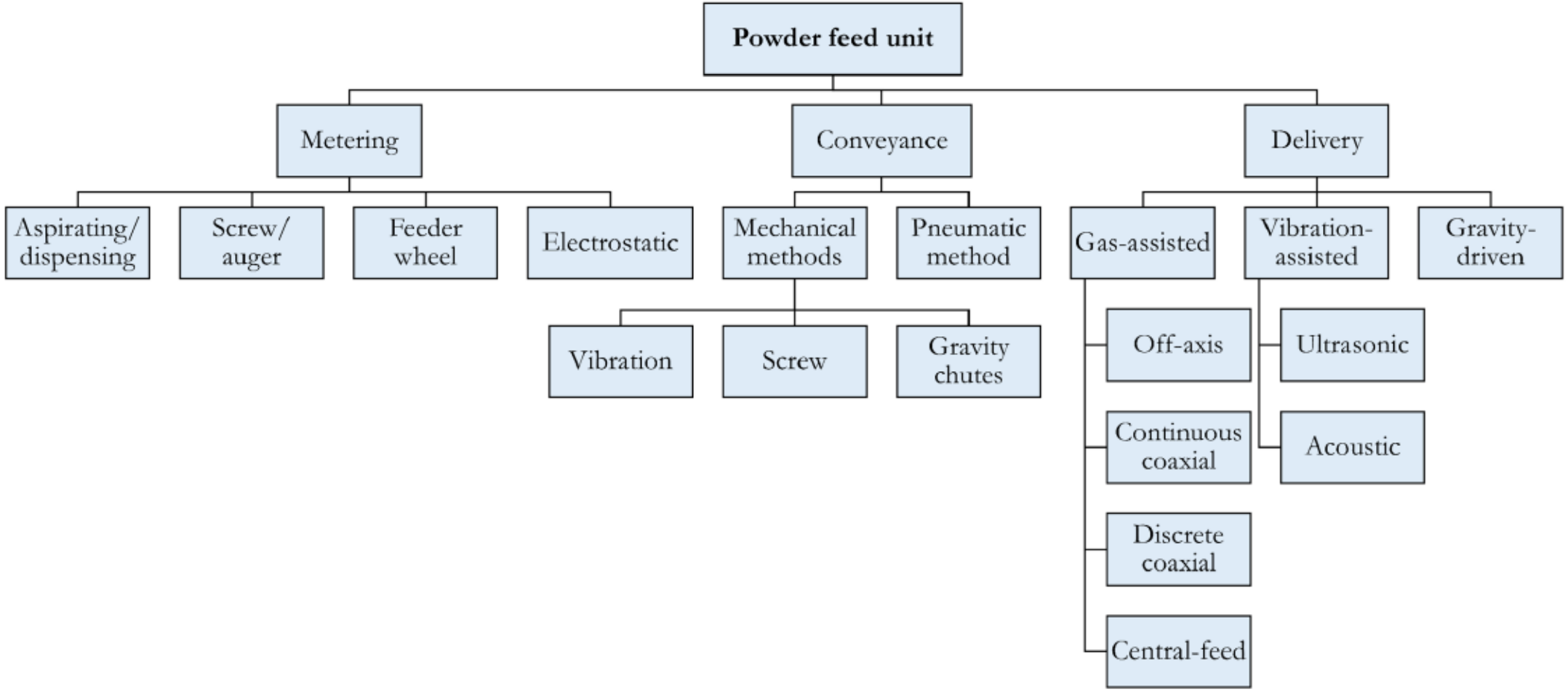
Directed Energy Deposition

Architecture of typical DED



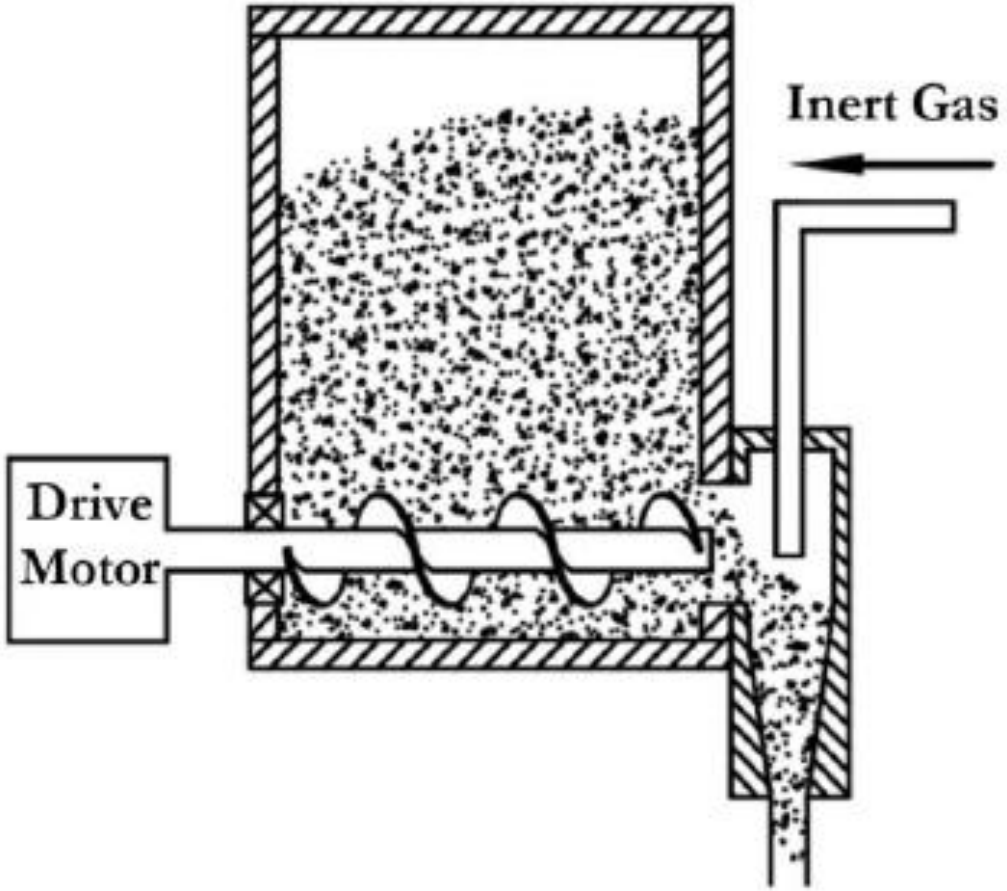
Directed Energy Deposition

Powder Feeding System



DED Powder Feeding System

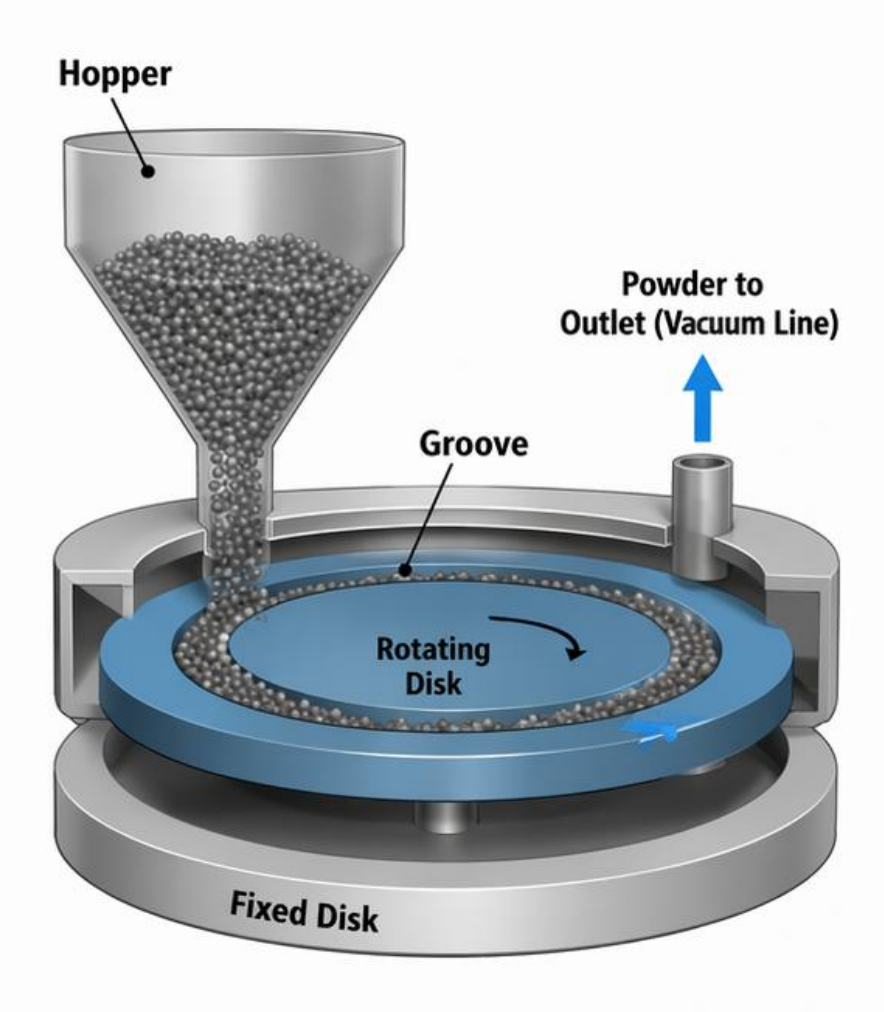
Screw / Auger type powder feeding





DED Powder Feeding System

Rotary Disk type

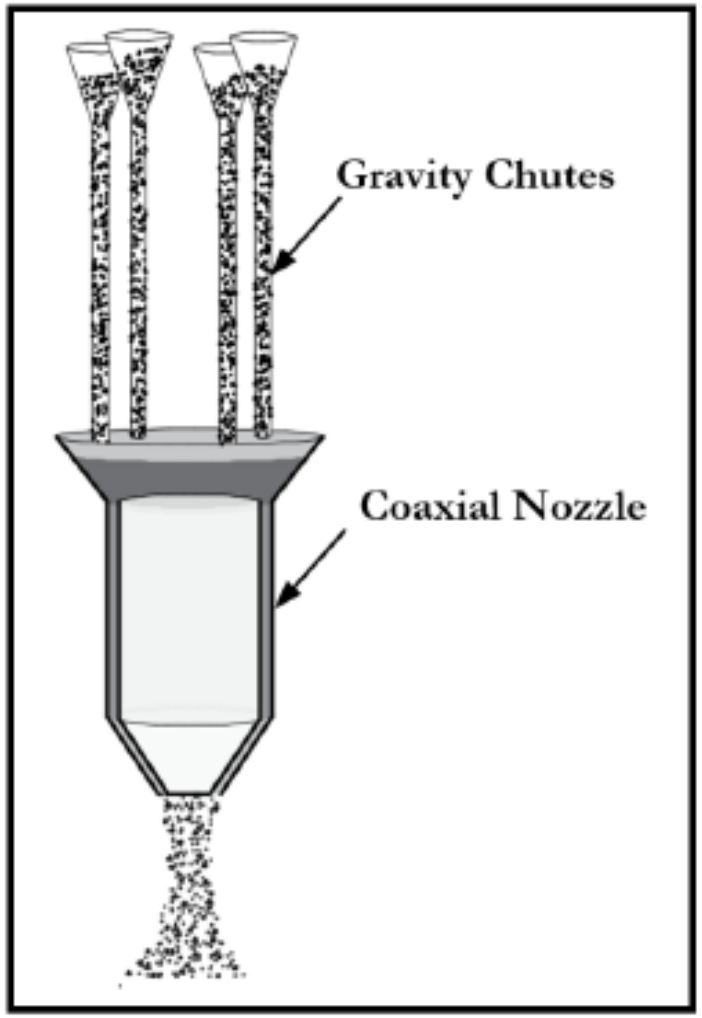


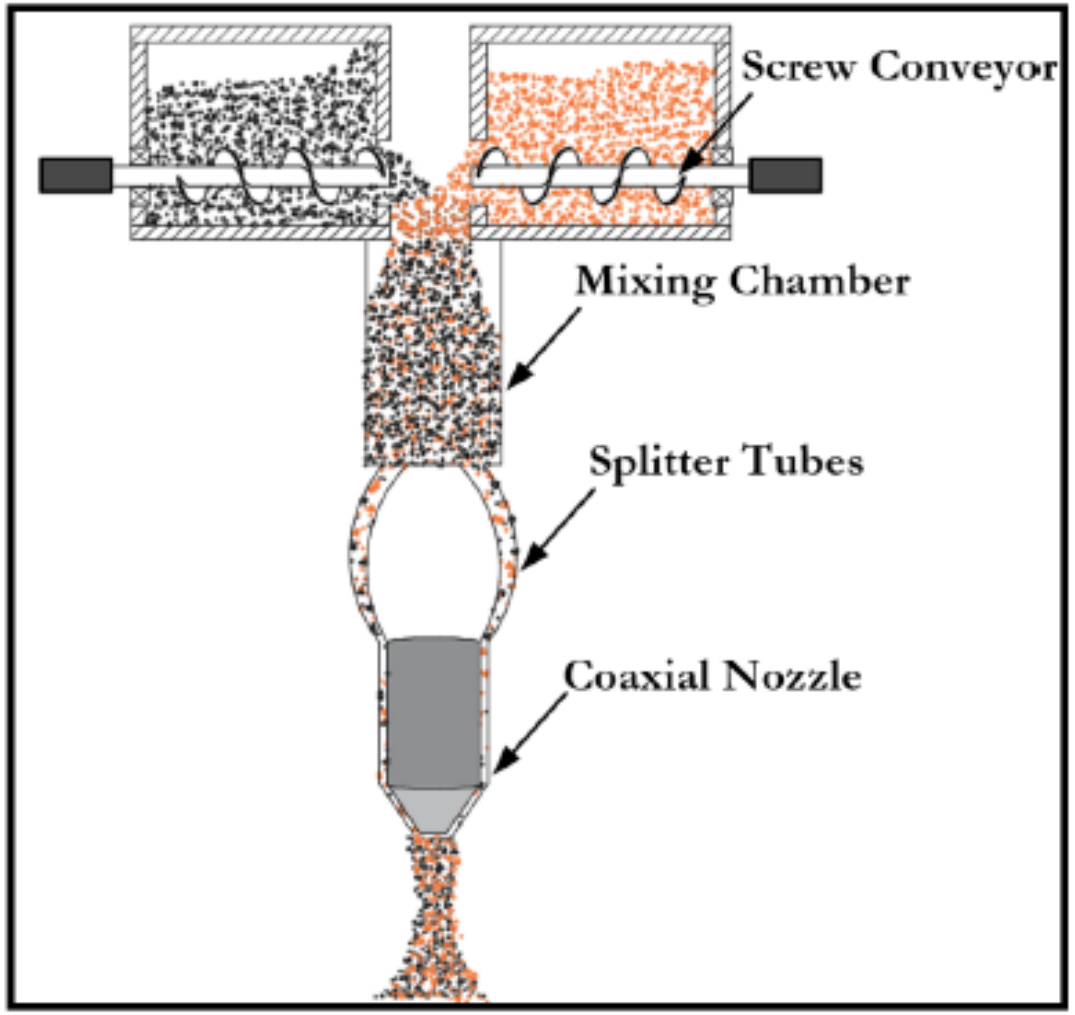
Flow Control
= Adjust RPM

Feed Rate (g/min)
= Volume per Rev (cm^3/rev)
× Bulk Density (g/cm^3)
× RPM

DED Powder Feeding System

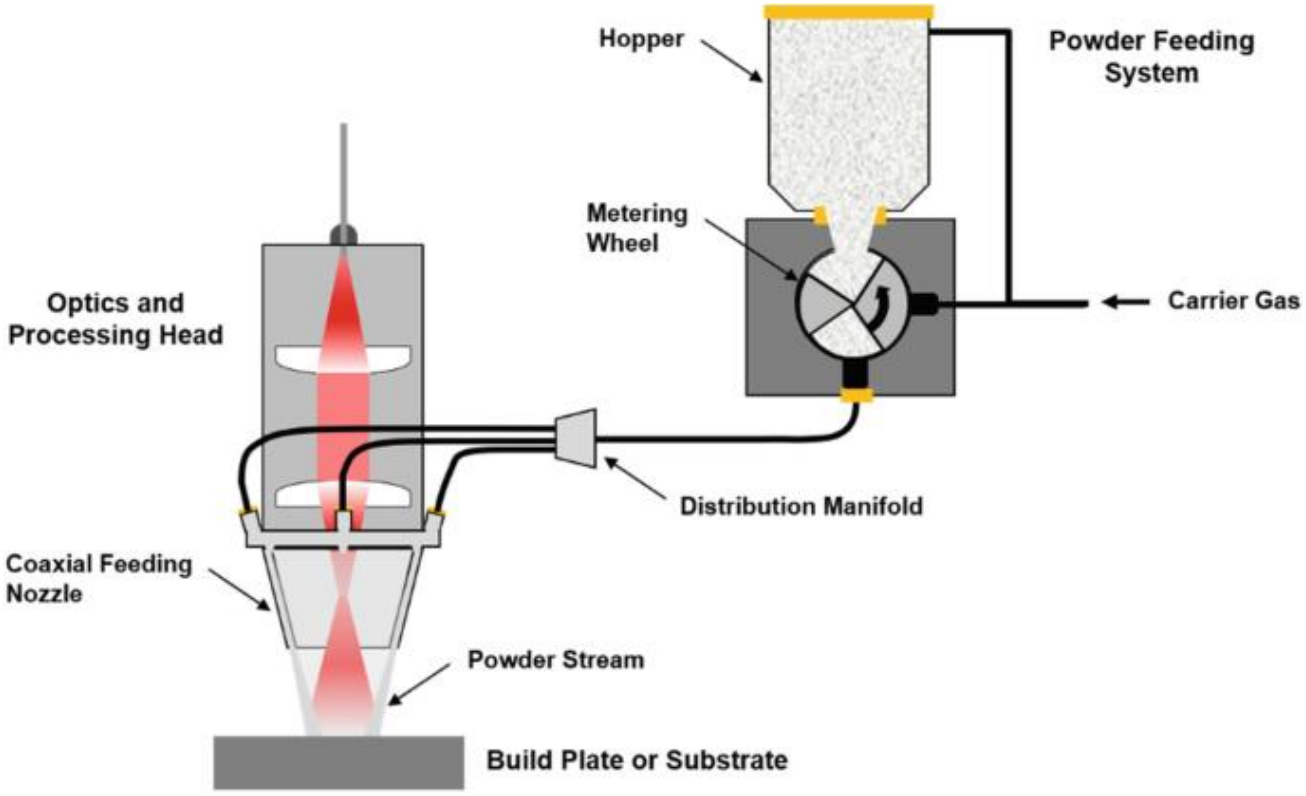
Gravity Based Powder feeding System

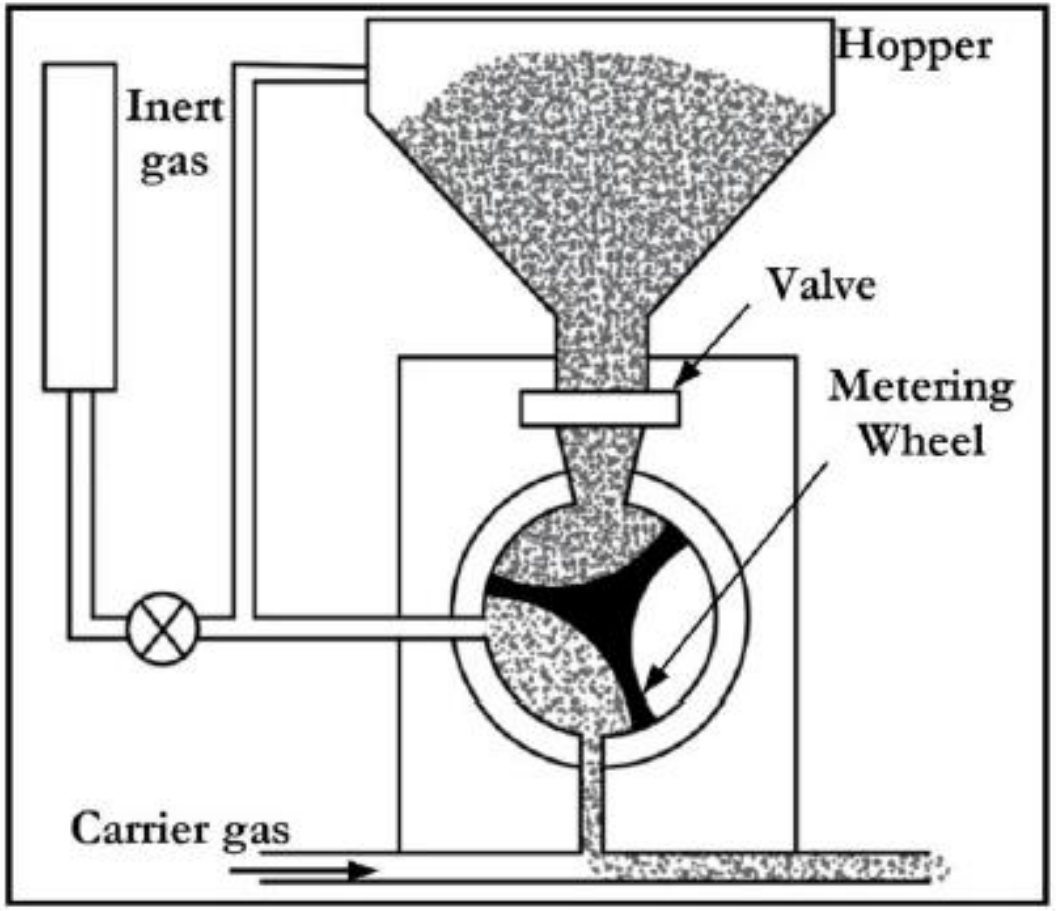




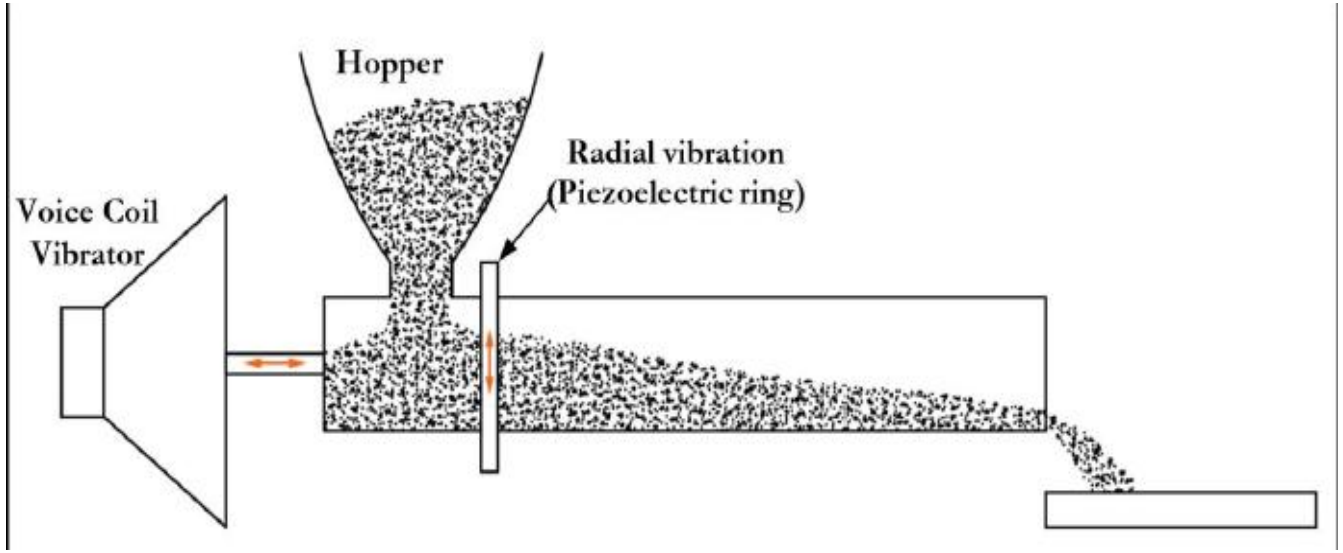
DED Powder Feeding System

Pneumatic type Powder Feeding System

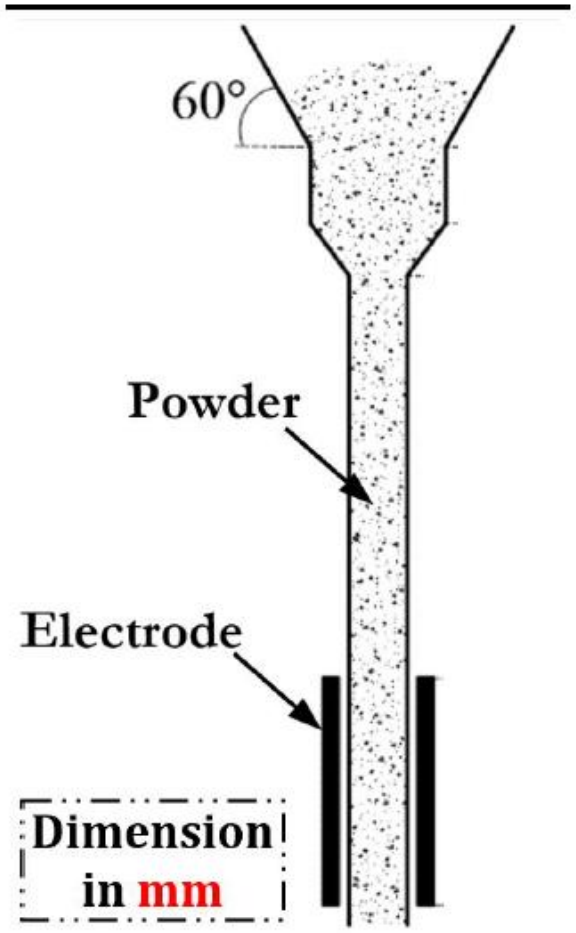




DED Powder Feeding System

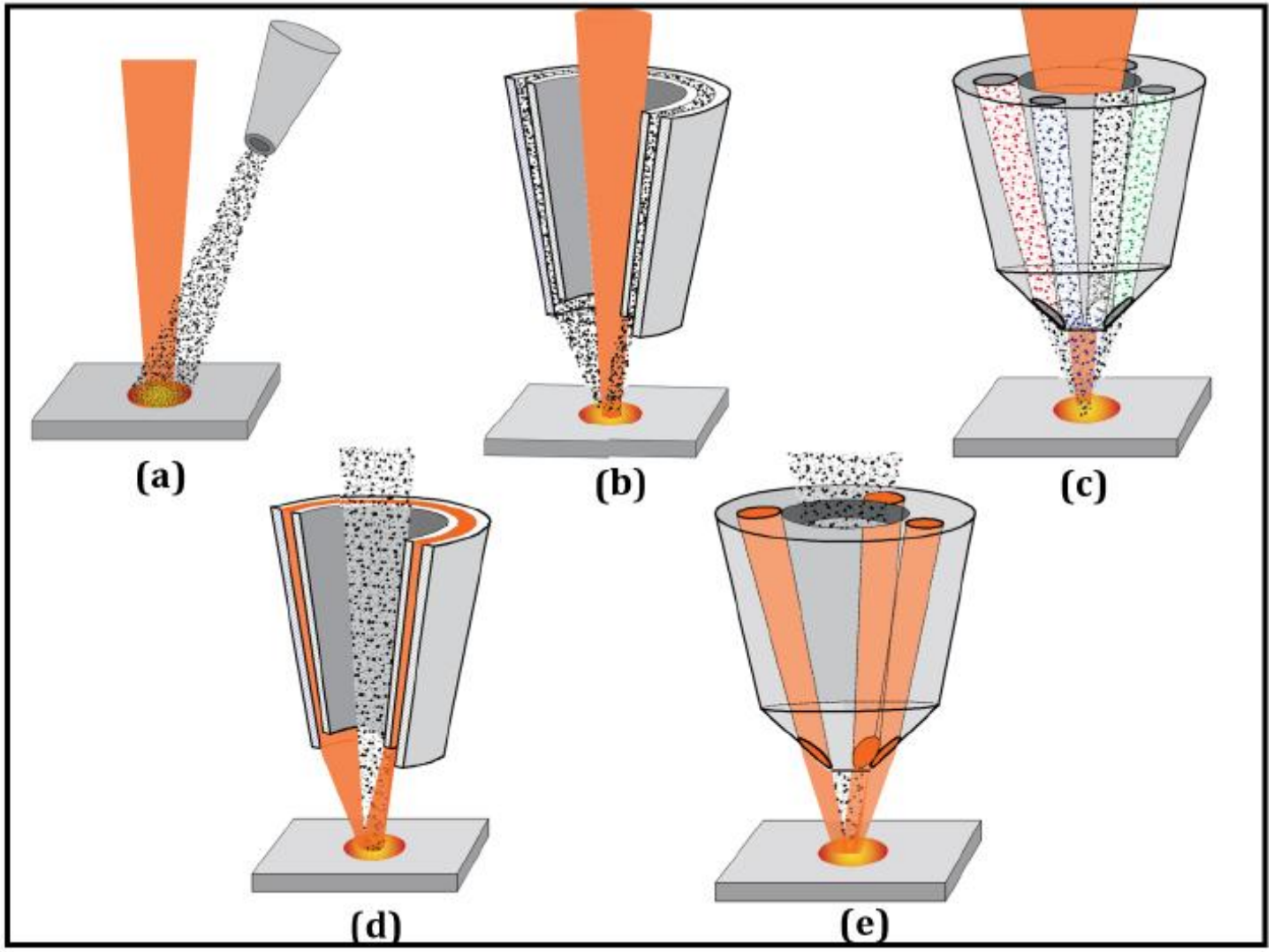


Vibratory Conveyance system



Electrostatic assisted system

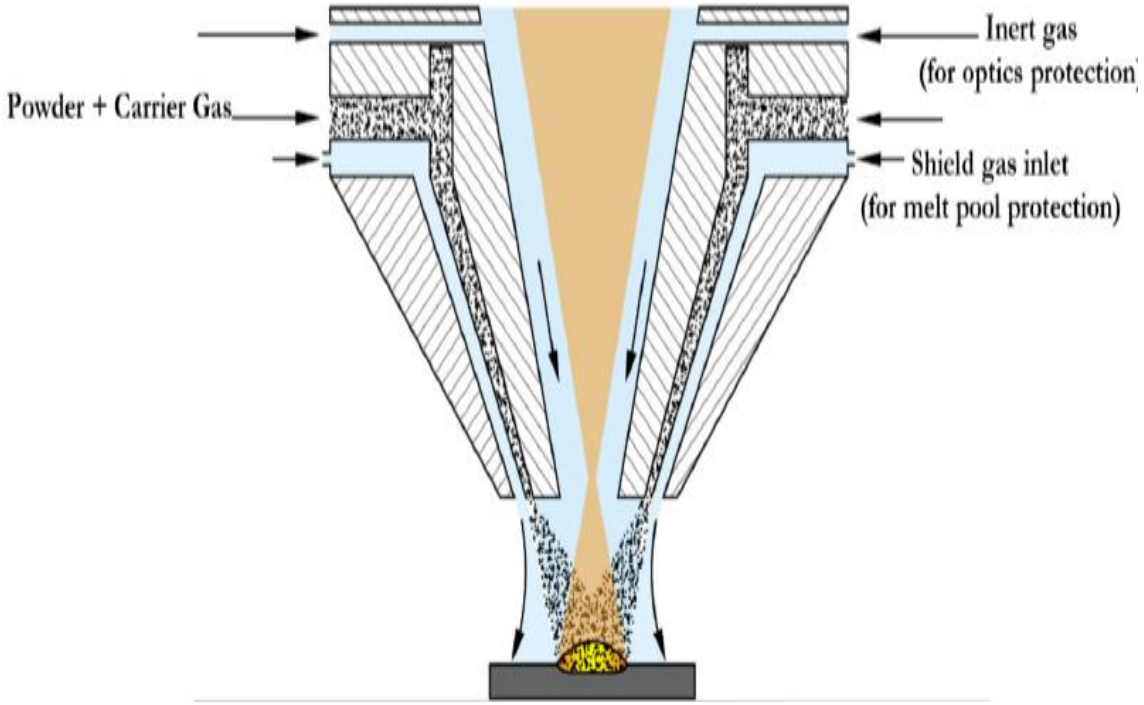
DED Powder Delivery Nozzle



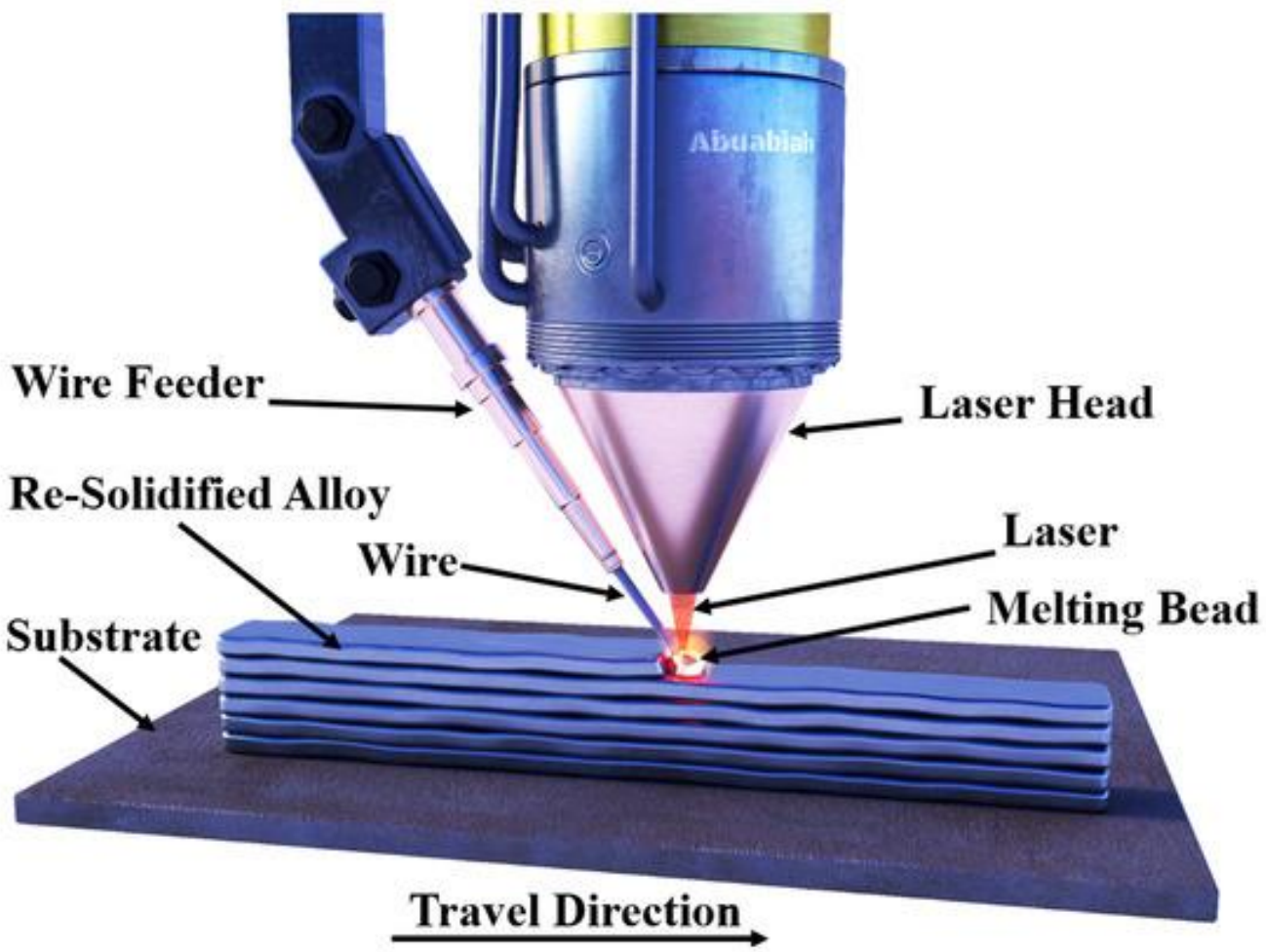
- ✓ Lateral Feed/ Off-Axis Nozzle
- ✓ Continuous Co-Axial
- ✓ Discrete Co-Axial
- ✓ Central Feed with Annular laser
- ✓ Central feed with discrete laser

DED : Powder Feed Type (Co-axial Nozzle Configuration)

Co-axial type (Common one)



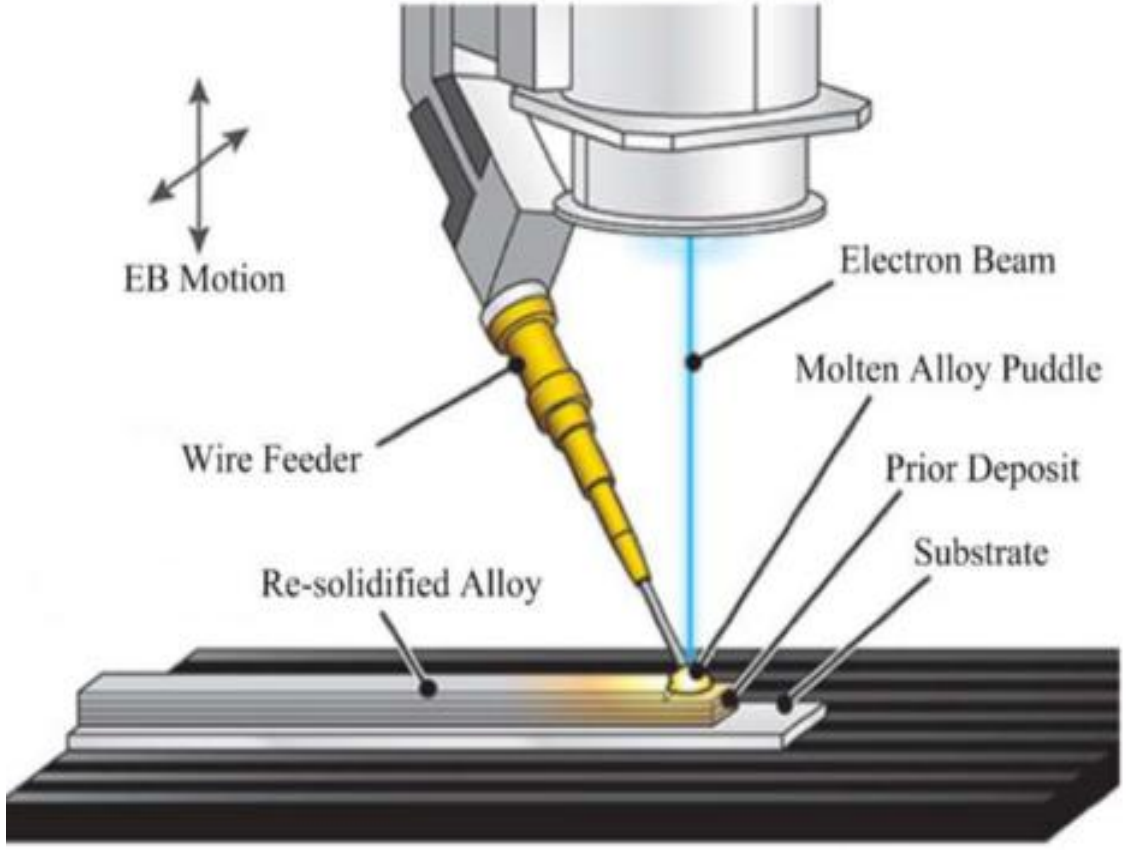
DED : Wire Feed Type



DED : Types of Energy Sources

Electron Beam (EB-DED)

Electron Beam Additive Manufacturing (EBAM)

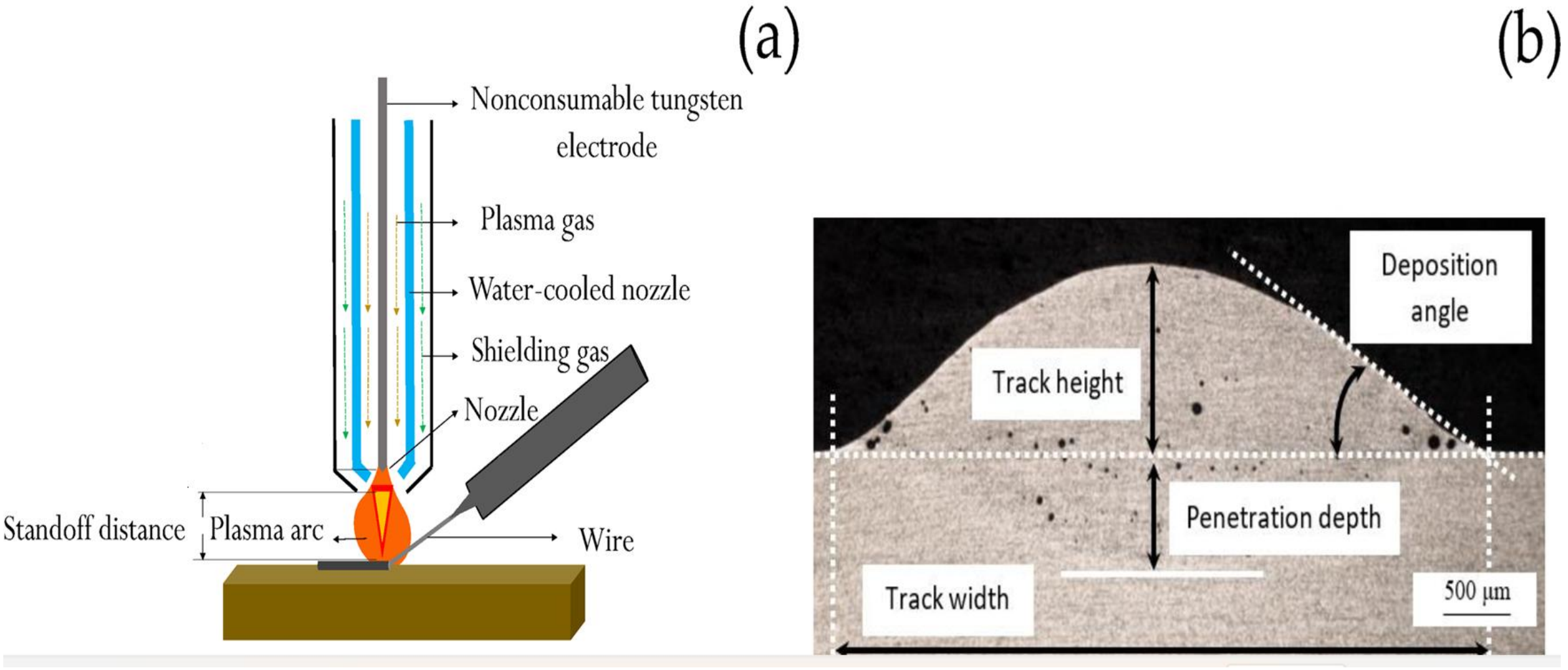




Model Rocket Engine- Copper Alloy
305 mm diameter and 485 mm height

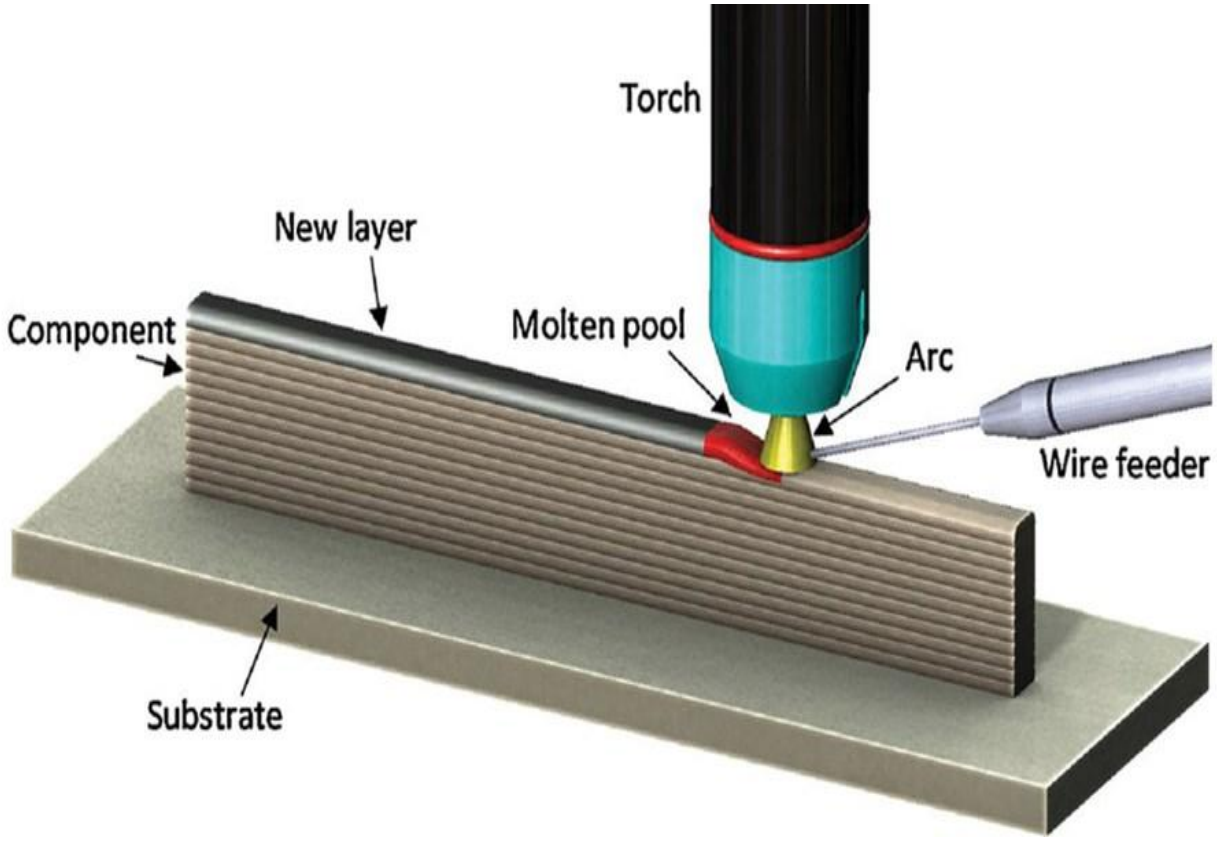
DED : Types of Energy Sources

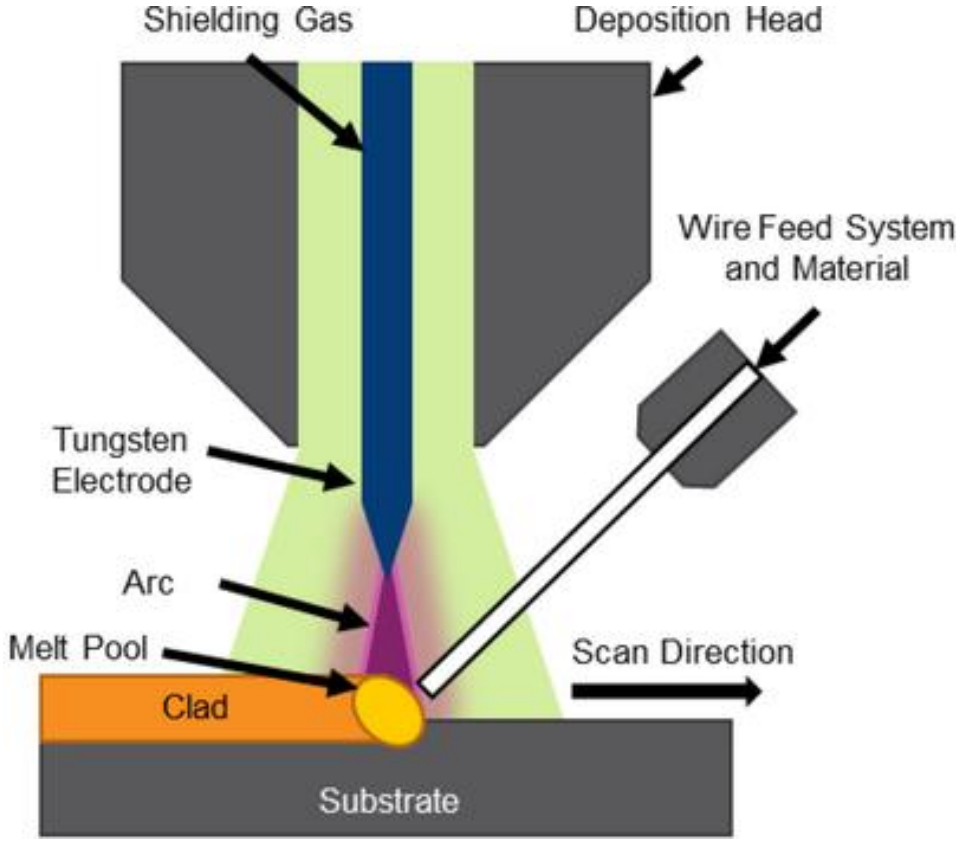
Plasma Transferred Arc (PTA-DED)



DED : Types of Energy Sources

Arc DED : Gas Tungsten Arc / Tungsten Inert Gas (GTAW-DED / TIG DED /MIG-DED)





Typical DED System

Typical LP- DED Machine Features

LP-DED

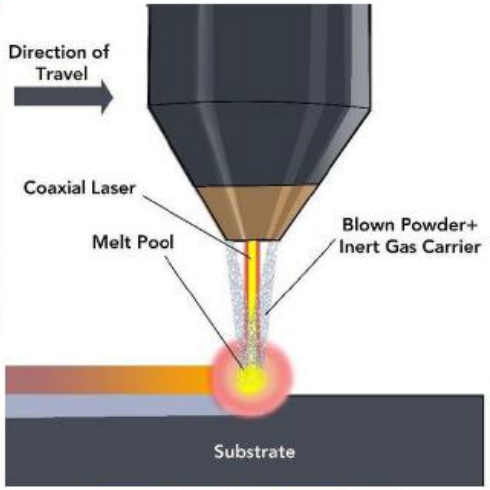
- Energy of a laser used to melt metal powders, deposit layer by layer
- Suitable for Manufacturing and/ or repair of parts using CAD data

Salient Features

- Build size (mm): Large Size up to 1.8m x 1.0m x 1.0m
- Payload: 800 Kg
- CNC controller, 5Axis
- Controlled atmosphere
- High power (2 kW) and low power 500 W laser heads
- Tool path software, Process monitoring system

Typical Components

- Impellers, exhaust casings, ducts for engines, Propellant tank components, end fittings, Gas bottles
- Development of FGMs, Multi-material products, Repairs



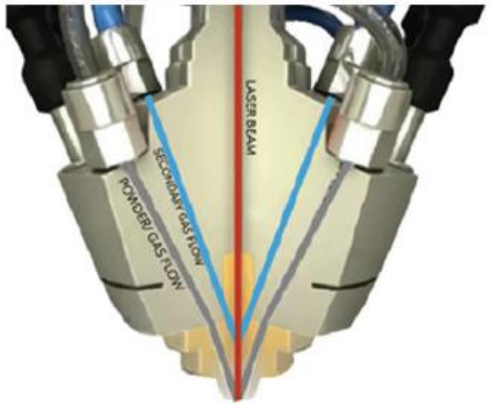


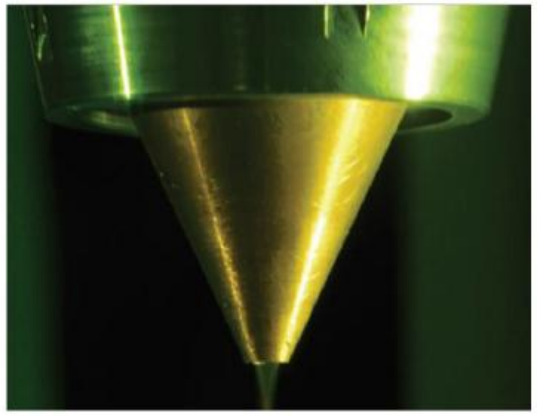


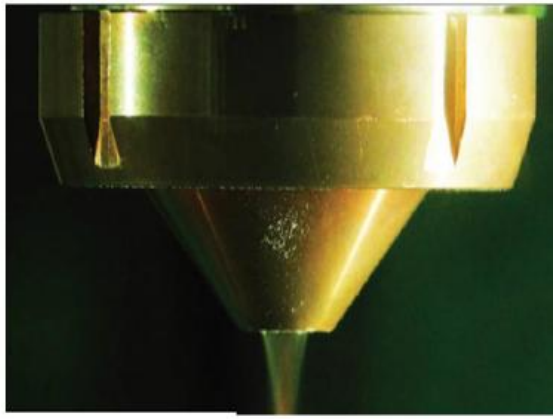
Courtesy: BeAM, France

DED : Process Parameters

- ✓ Laser Power
- ✓ Scanning Speed
- ✓ Powder / Wire Feed Rate
- ✓ Spot Size
- ✓ Substrate Temperature
- ✓ Carrier/Shielding Gas
- ✓ Layer Thickness
- ✓ Hatch Spacing
- ✓ Stand-off distance



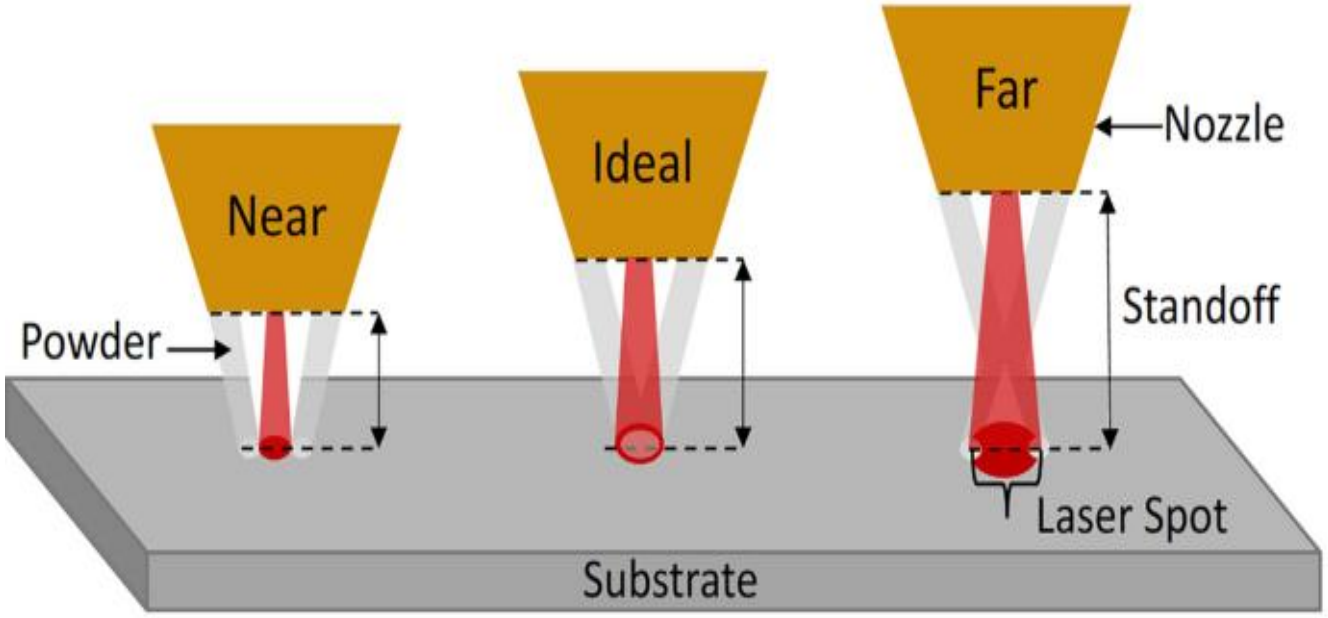


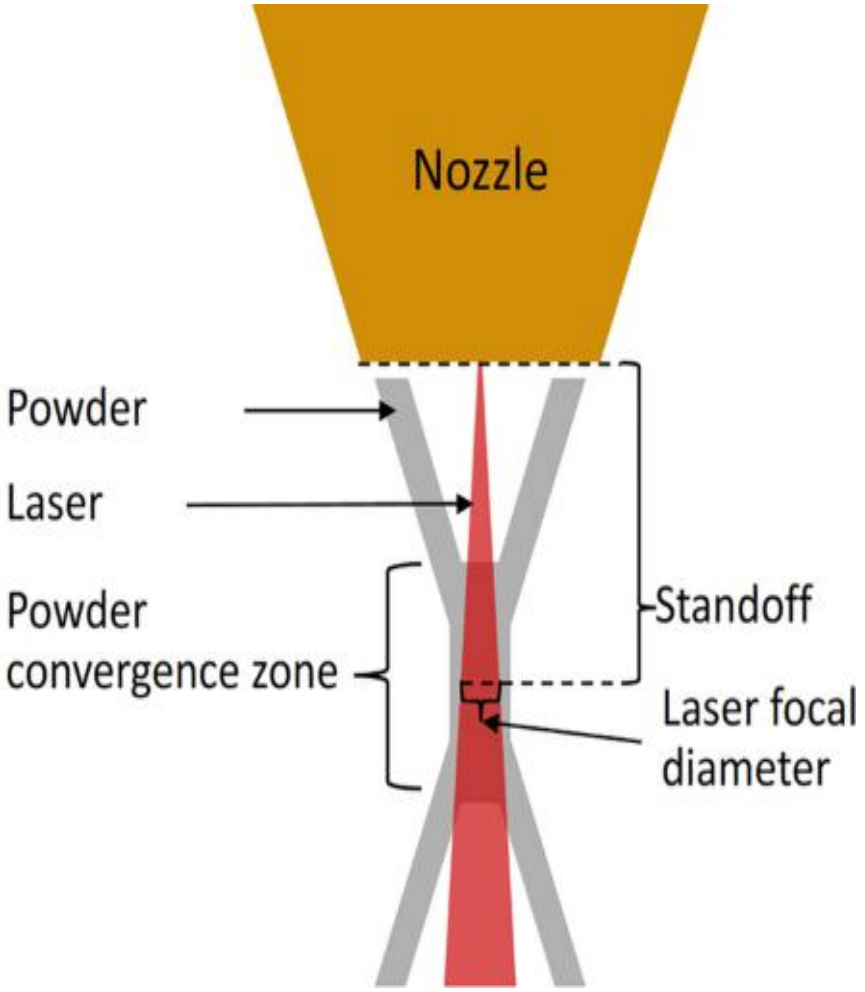


Typical Process Parameters in DED

Laser	**Power, Beam Diameter, Beam Profile, Working Distance, Wavelength, etc.**
Feeding	Nozzle Type, Powder Flow Rate, Gas Flow Rate
Material & Design	Material of Powder, Diameter Distribution of Powder, Shape of Powders, Material of Substrate, Heat Treatment of Substrate, Geometry of Substrate, Geometry of Deposited Region
Motion & Path	Scan Speed, Track Overlap Ratio, Deposition Path, Interlayer Dwell Time, Preheating Temperature

DED : Standoff distance





DED : Energy Parameters

Energy Per Unit Length (EL)

Linear Energy Density (LED)

$$EL = \frac{Laser\ Power\ (P_L)}{Scanning\ Speed\ (V)}$$ J/ mm

Defines the amount of energy deposited along the scanning path

Mass Per Unit Length / Speed Ratio

Material Feed Per Unit Traverse Length

$$MFL = \frac{Material\ Feed\ Rate\ (\dot{m})}{Scanning\ Speed\ (V)}$$

Material feed rate refers to Wire feed or Powder Feed

DED : Energy Parameters

A suitable combination of energy per unit length and speed ratio is crucial for a defect-free process.

- ✓ It directly impacts the thermal gradient and cooling rates, influencing grain structure (columnar vs. equiaxed).
- ✓ Increasing the energy per unit length typically increases the track width and height, resulting in a larger bead area.
- ✓ Lower scanning speeds (which increase LEL) decrease cooling rates, resulting in coarser, columnar, or dendritic microstructures.
- ✓ Too high an energy input can lead to keyholes, excessive dilution and cracking.
- ✓ Too low energy leads to lack of fusion and porosity.

DED : Energy Parameters

While energy density controls how much space the laser carves out, material feed controls how much that space is filled.

Increasing the material feed per unit length directly leads to higher bead (track) height and width . More mass is captured and solidified within the melt pool per unit of travel.

If the material feed is too high relative to the available energy, catchment efficiency may drop because there isn't enough heat to melt all the incoming powder or wire.

What is catchment efficiency ??

Increasing the material feed rate per unit length can reduce the melt pool's temperature gradient, which often facilitates the formation of finer grains due to a faster transition from liquid to solid state at the pool's boundaries

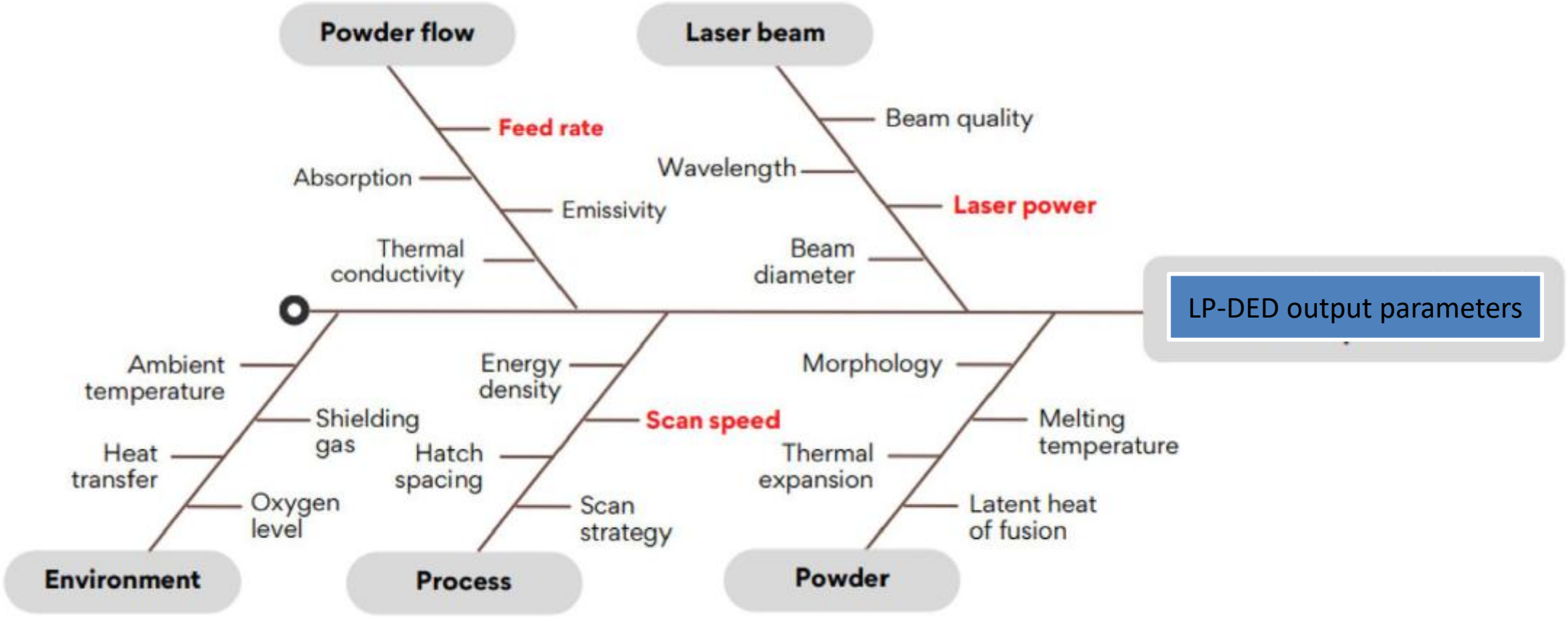
DED : Catchment Efficiency

Catchment efficiency

Catchment Efficiency in Directed Energy Deposition (DED) is the ratio of the mass of material effectively deposited into the part to the total mass of feedstock supplied through the nozzle.

It is a critical metric for process viability, as low efficiency leads to significant material waste and increased

Laser Based Powder DED : Process Variables

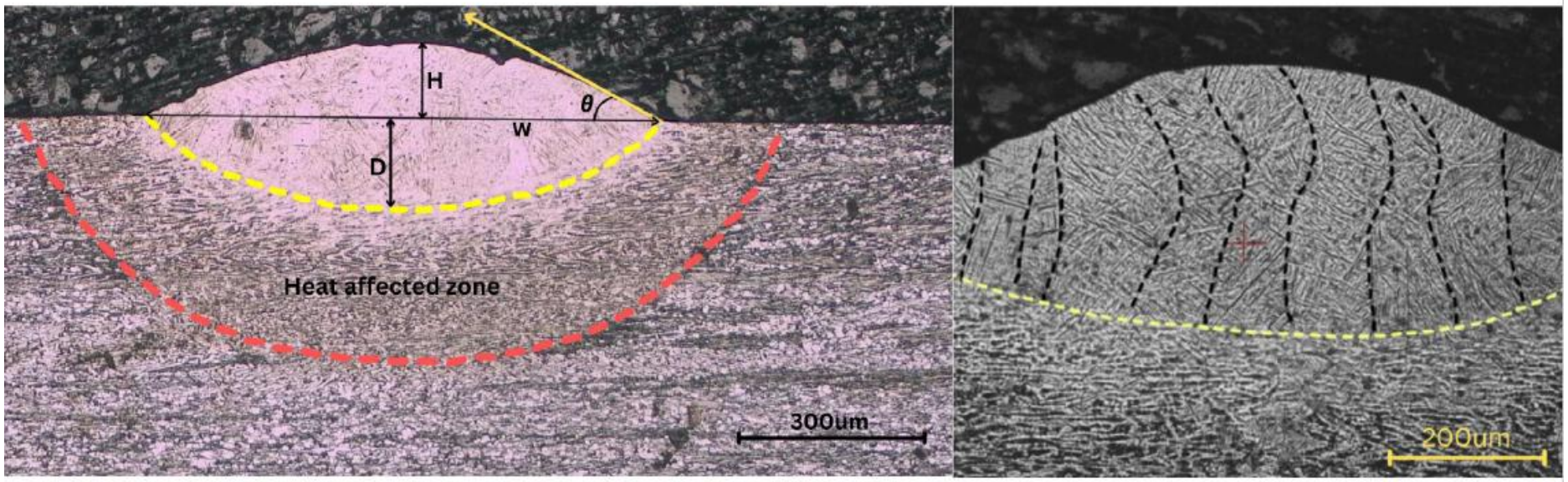


Laser Based Powder DED : Process Optimization

$$\text{Linear energy density } (\text{Jmm}^{-1}) = \frac{\text{laser power (W)}}{\text{speed } (\text{mms}^{-1})}$$

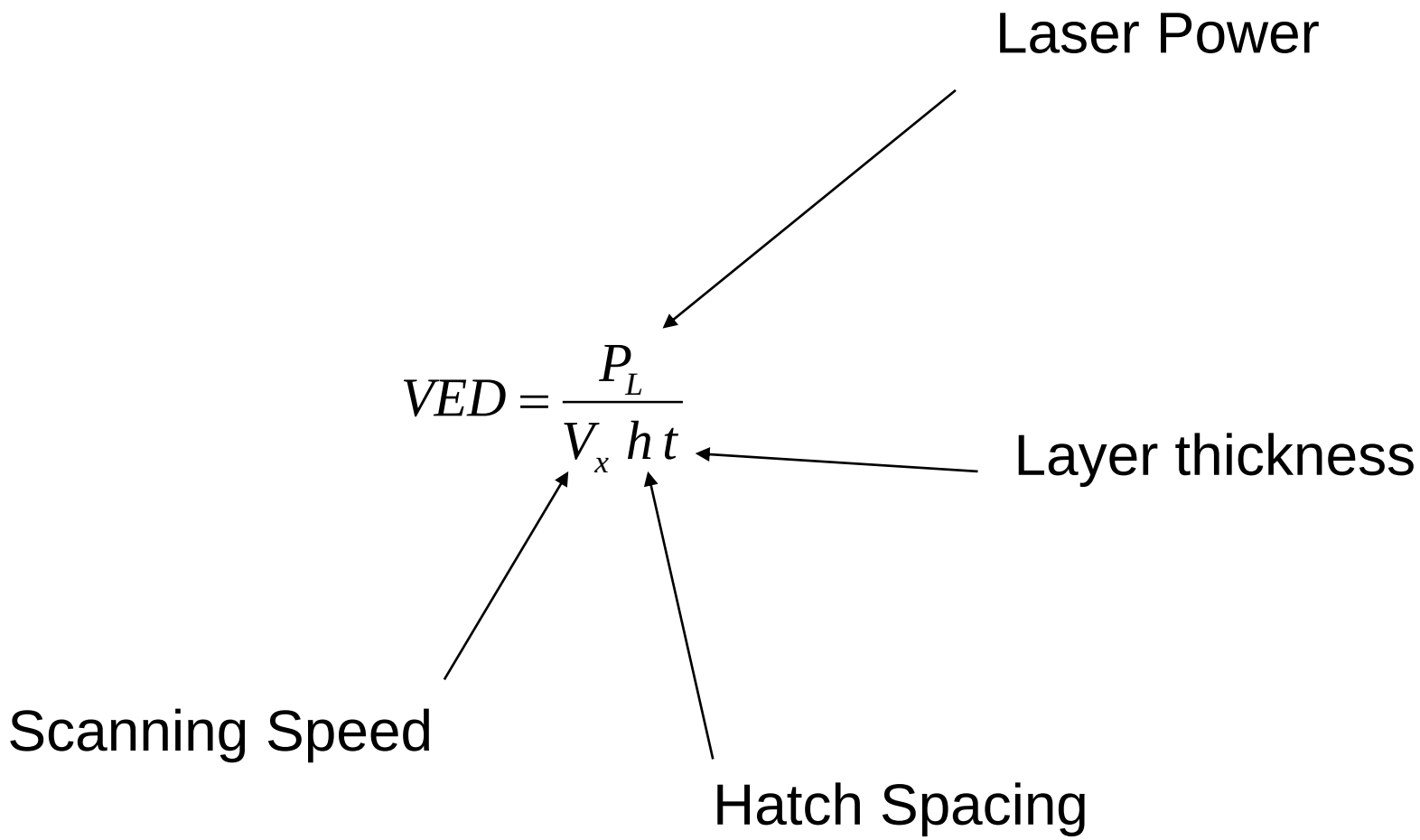
$$\text{Specific energy density } (\text{Jmm}^{-1}) = \frac{\text{laser power (W)}}{\text{speed } (\text{mms}^{-1}) \times \text{beam diameter (mm)}}$$

$$\text{Dilution}(\%) = \frac{\text{penetration depth(D)}}{\text{height(H)} + \text{penetration depth(D)}} \times 100$$



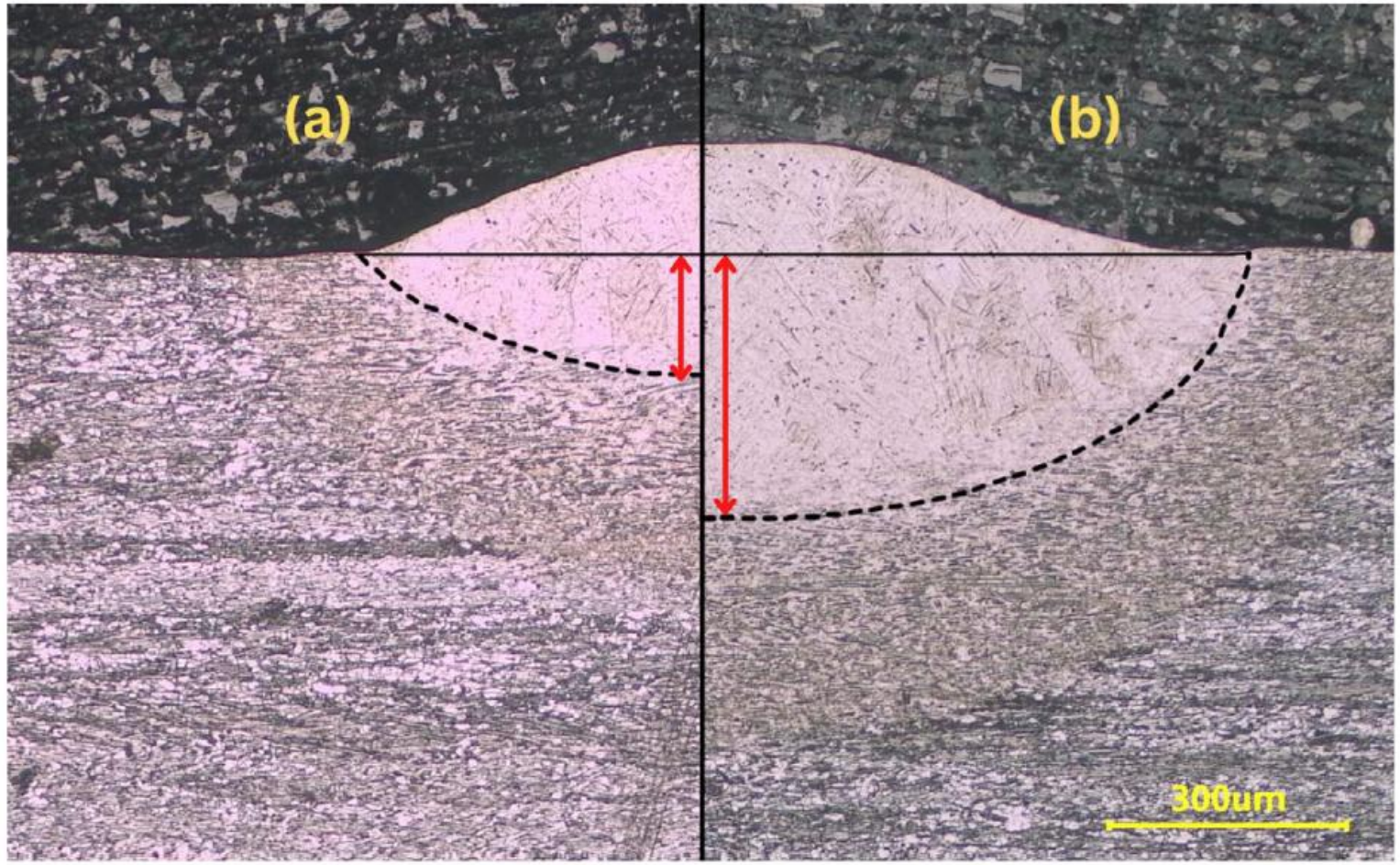
Formula for Dilution is approximated for simple/ single track

The volumetric energy density in DED



Laser Based Powder DED : Single Track Study (Case Study)

Single Track Studies

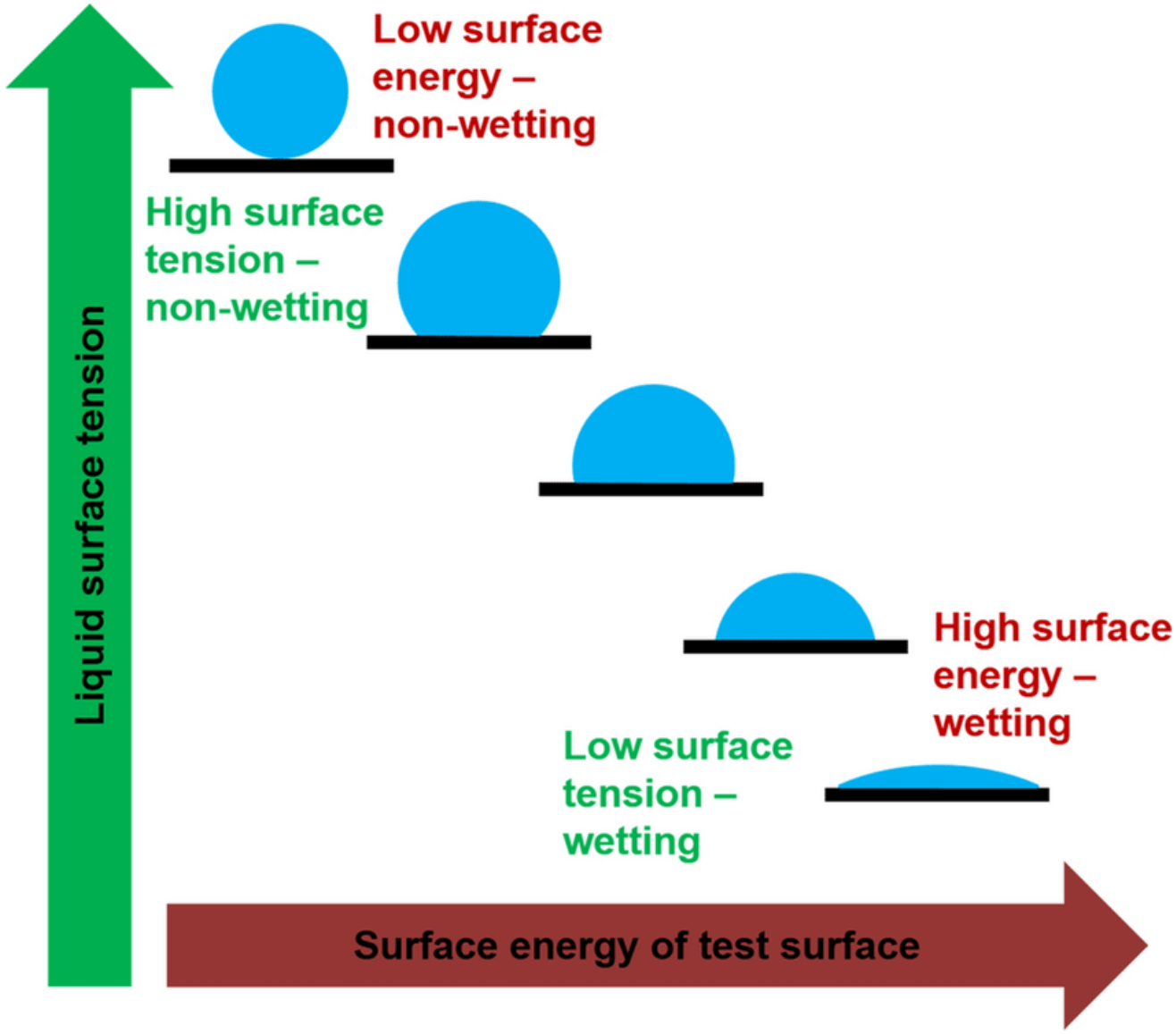


Optical micrographs of cross-sectional views of single-tracks deposited using LDED with (a) 300 W, 720 mm min−1; (b) 700 W, 960 mm min−1. In both of the scenarios

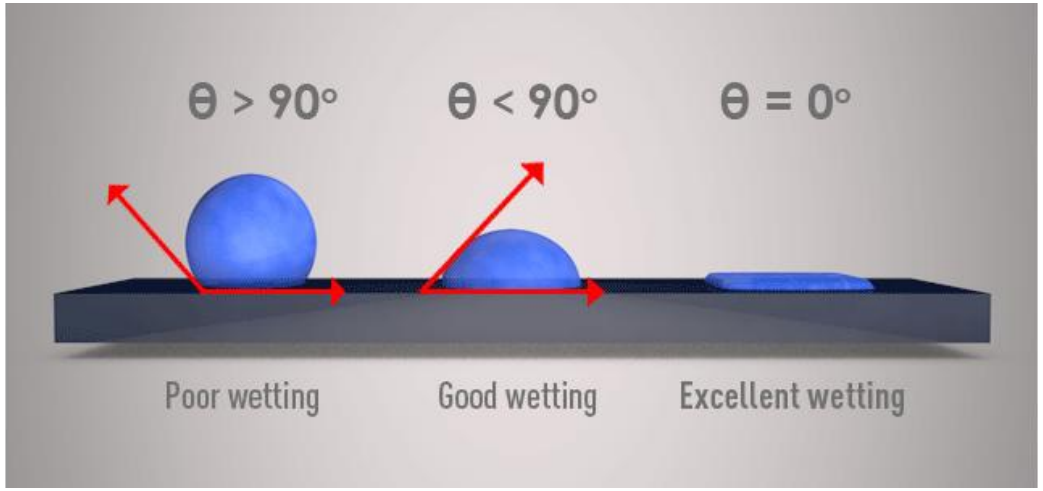
Optical micrographs of cross-sectional views of single tracks deposited using LP-DED with (a) 300 W, 720 mm/min; (b) 700 W, 960 mm/min.

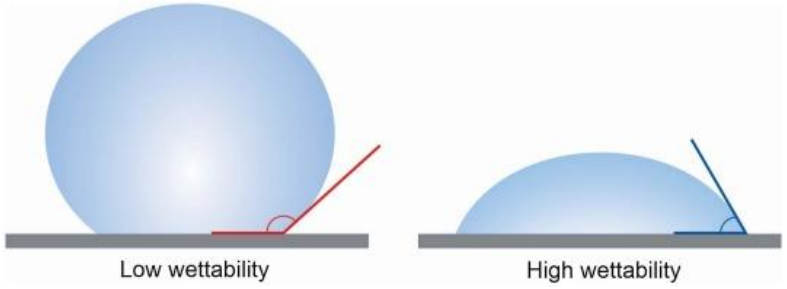
In both cases, bead heights are same, but depth of penetration is higher in (b), due to high energy density input

Laser Based Powder DED : Wetting of Molten Liquid



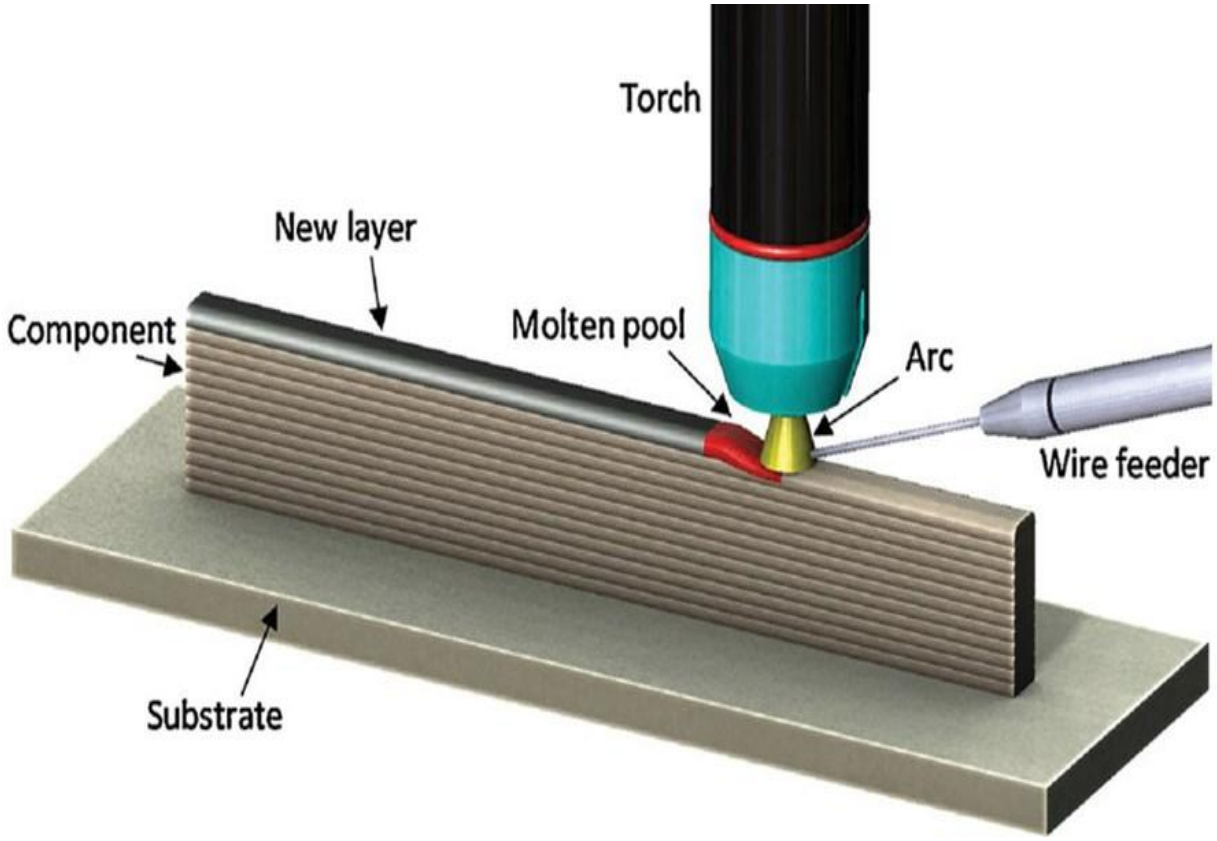
Wetting refers to the ability of the molten pool to spread over and adhere to the substrate surface.

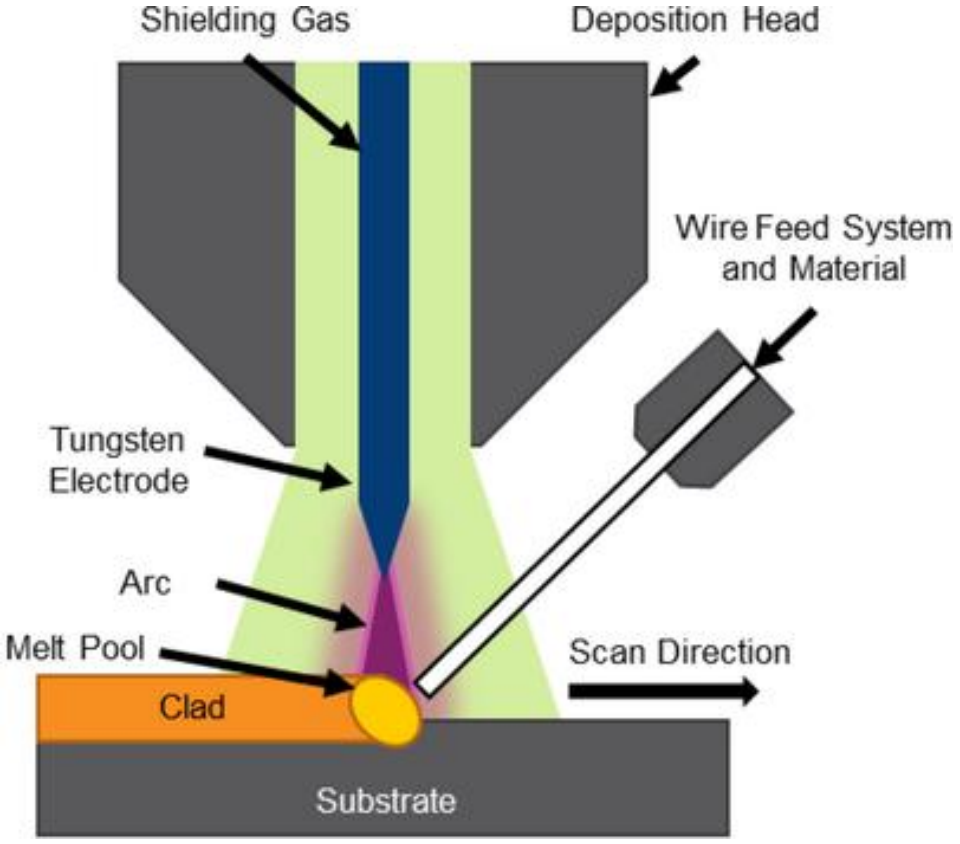




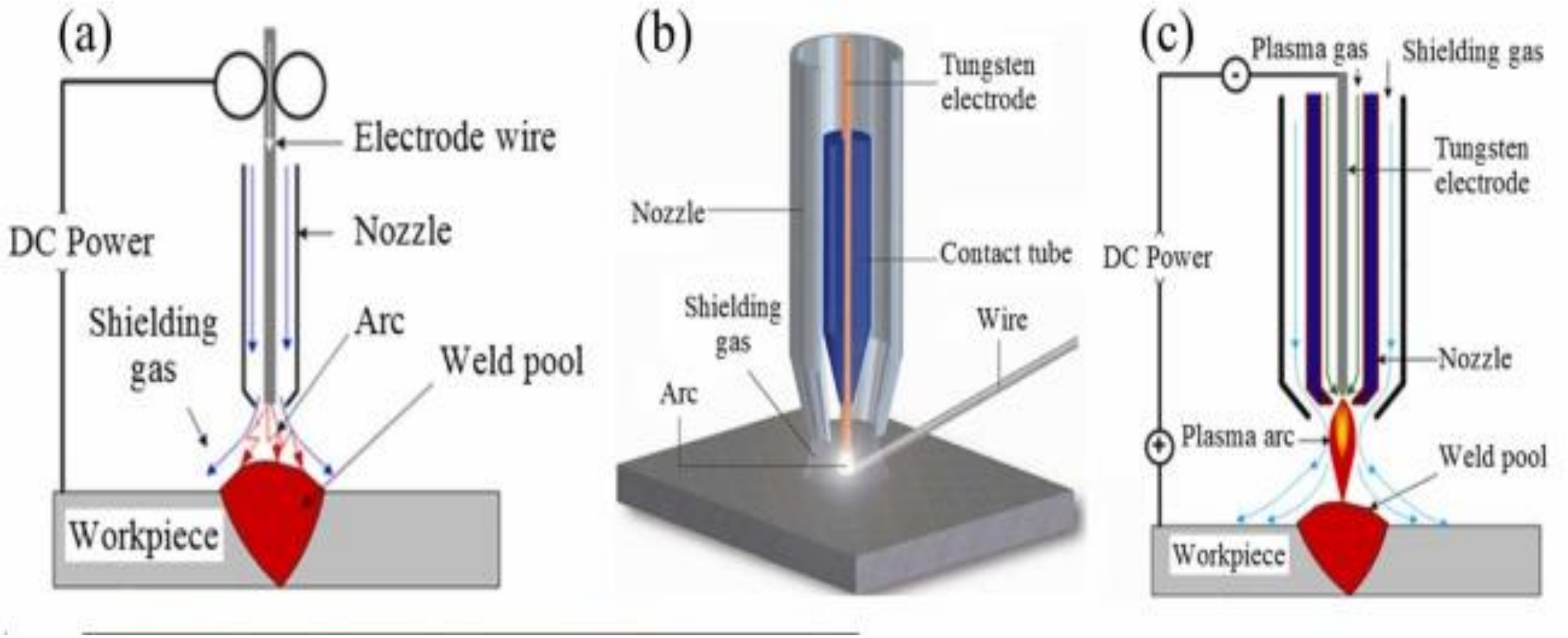
Wire Arc Additive Manufacturing (WAAM)

Arc is used to melt the wire and deposit



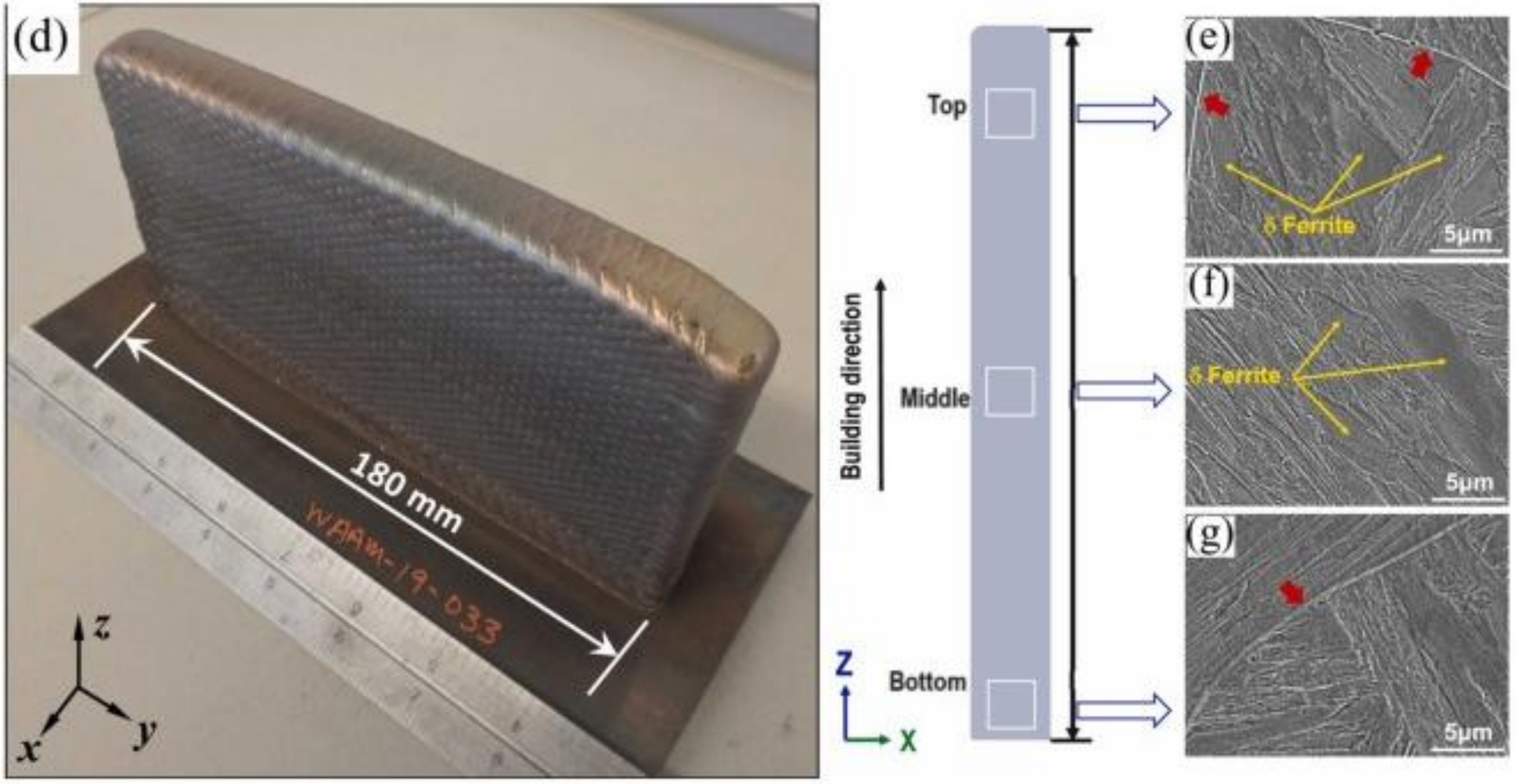


Wire Arc Additive Manufacturing (WAAM)



MIG based System *TIG based System* *Plasma Arc based System*

Wire Arc Additive Manufacturing (WAAM)



Case Study : Microstructure at top, middle and bottom locations of WAAM P91 steels

Wire Arc Additive Manufacturing (WAAM)

Major Components

Power Source

DC Power Source

AC Power Source

Generally DC preferred due to the control of heat input offered

Wire Feed System

Motion Control System : CNC or Robotic System

Weld Torch

Auxiliary System : Shielding Gas System , Pre-heating / Cooling systems if any

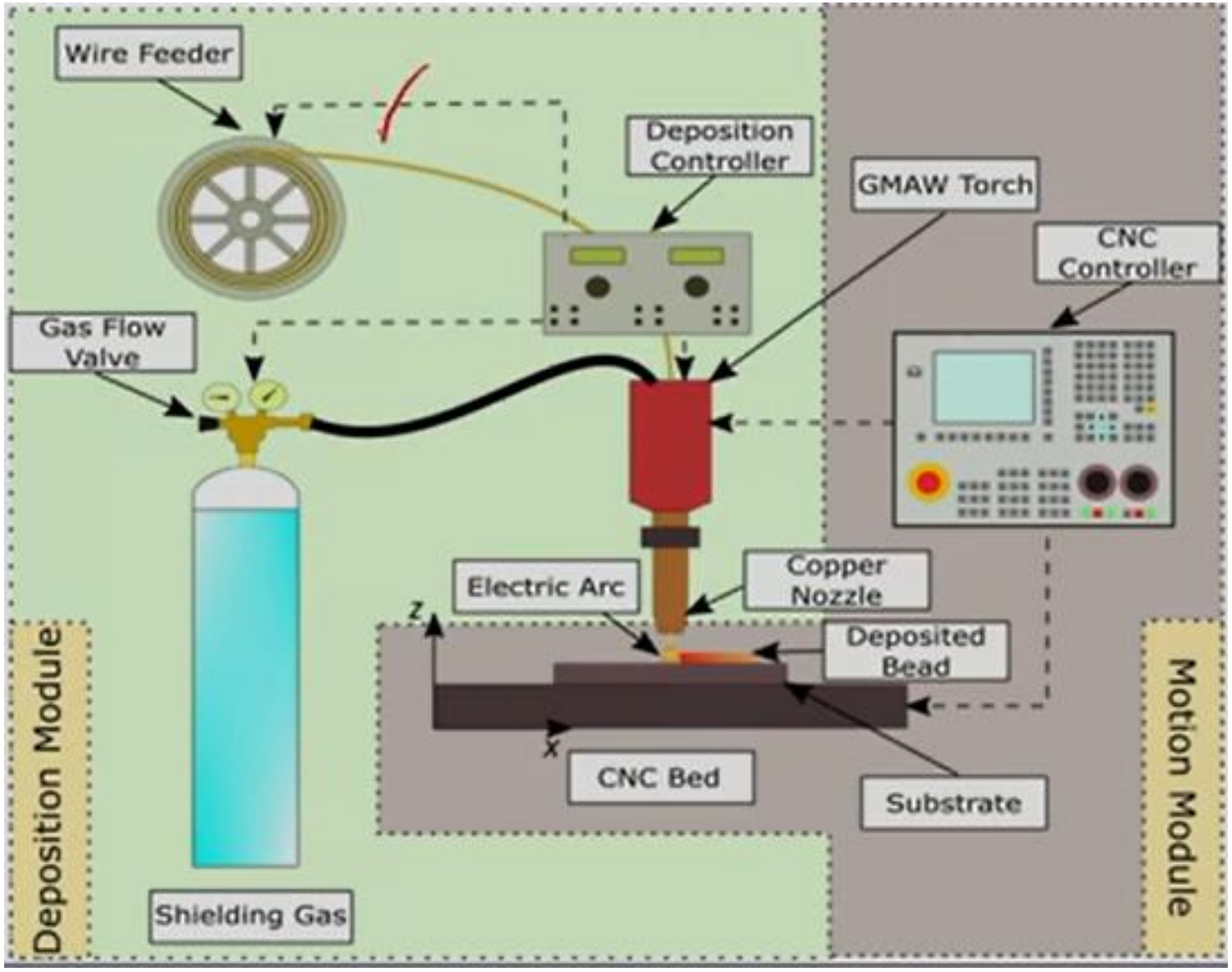
Wire Arc Additive Manufacturing (WAAM)

WAAM

GMAW (MIG)

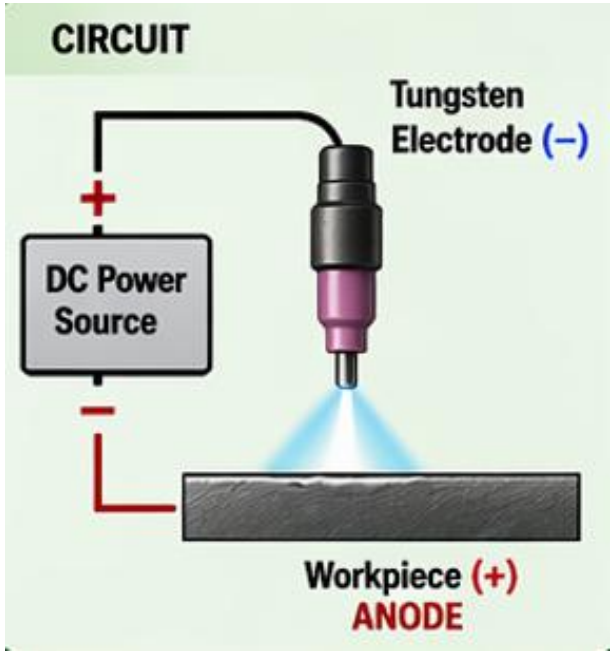
TIG

Based systems

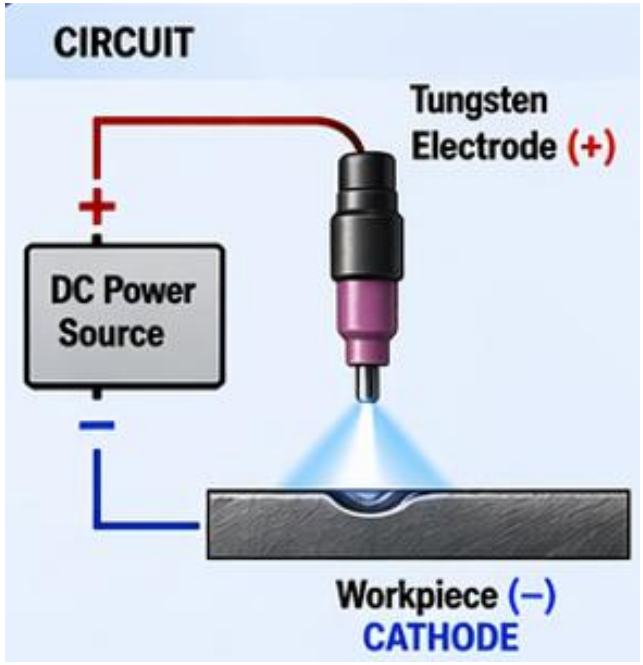


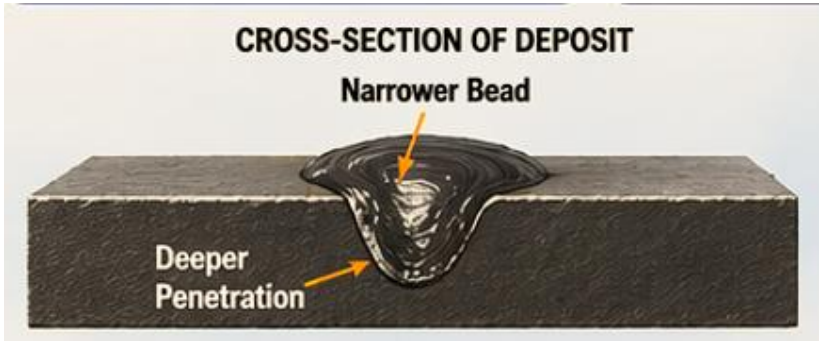
WAAM : MIG / TIG system

How polarity control heat, metal transfer and bead Shape

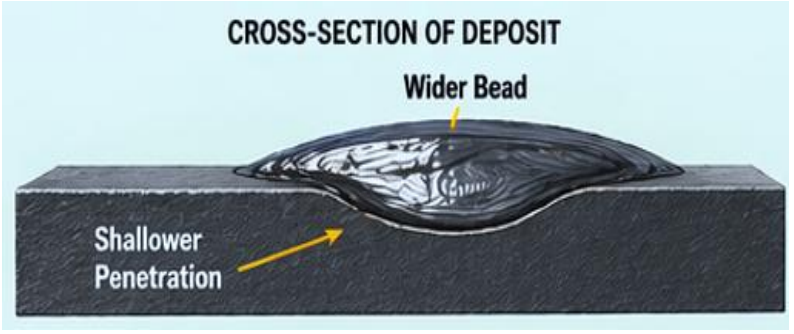








DCSP Vs DCRP

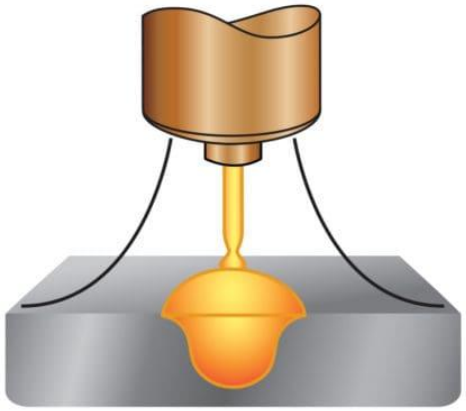


Wire Arc Additive Manufacturing (WAAM)

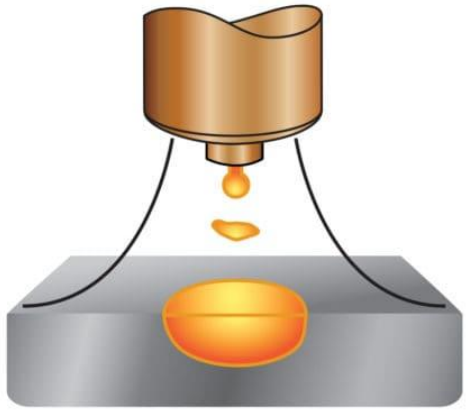
WAAM : MIG/ TIG System

Mode of Metal Transfer

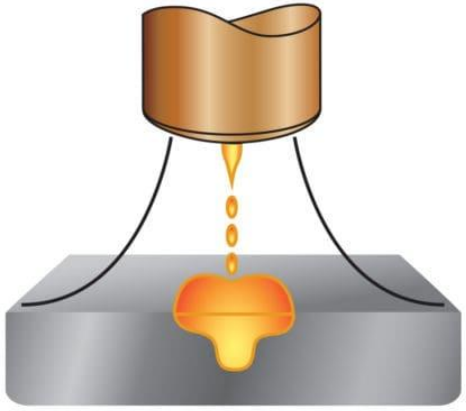
1. Short Circuit Transfer
2. Globular Transfer
3. Spray Transfer
4. Pulsed Metal Transfer



Short Circuit Transfer



Globular Transfer



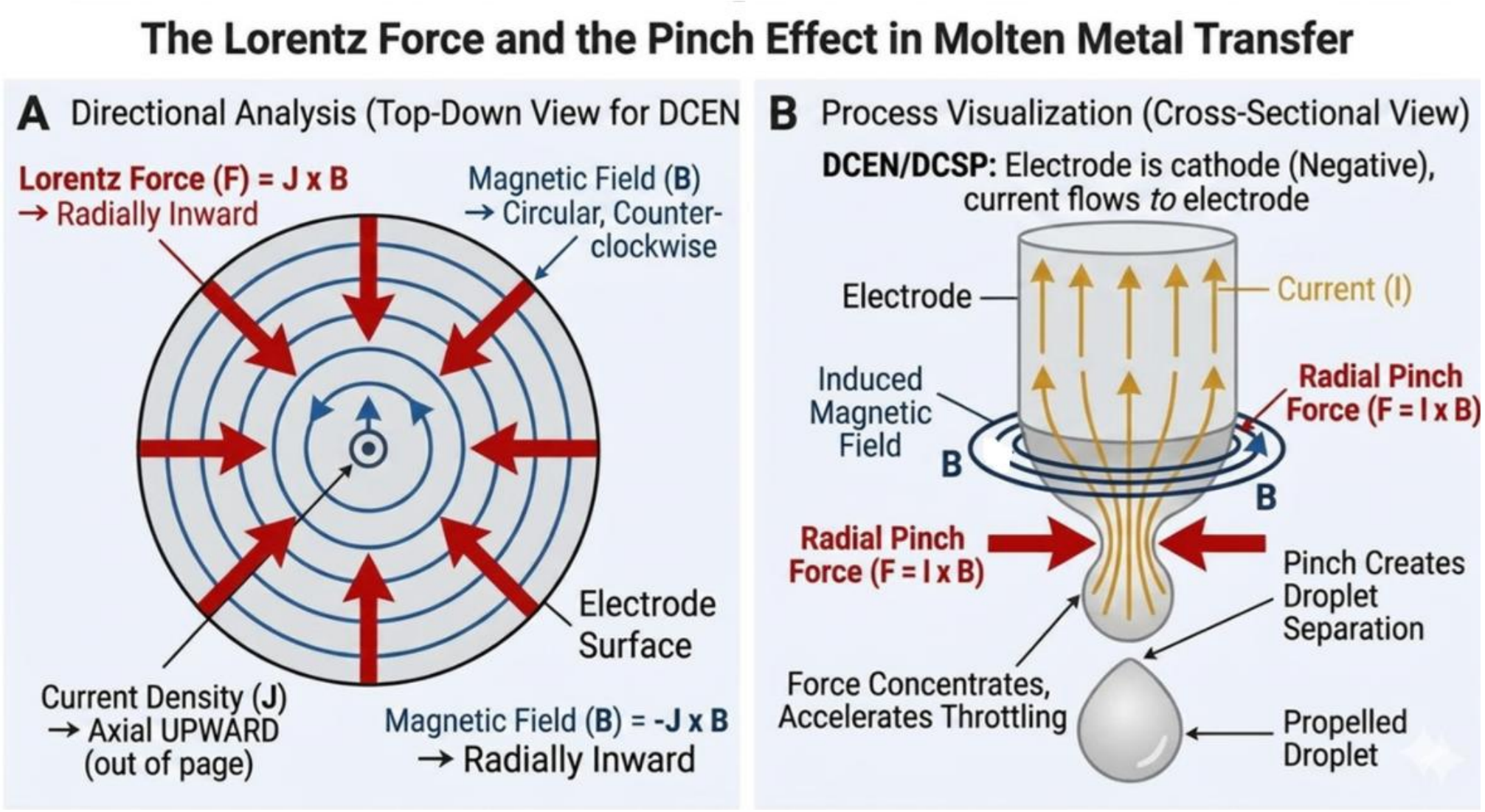
Spray Transfer



Pulse Spray Transfer

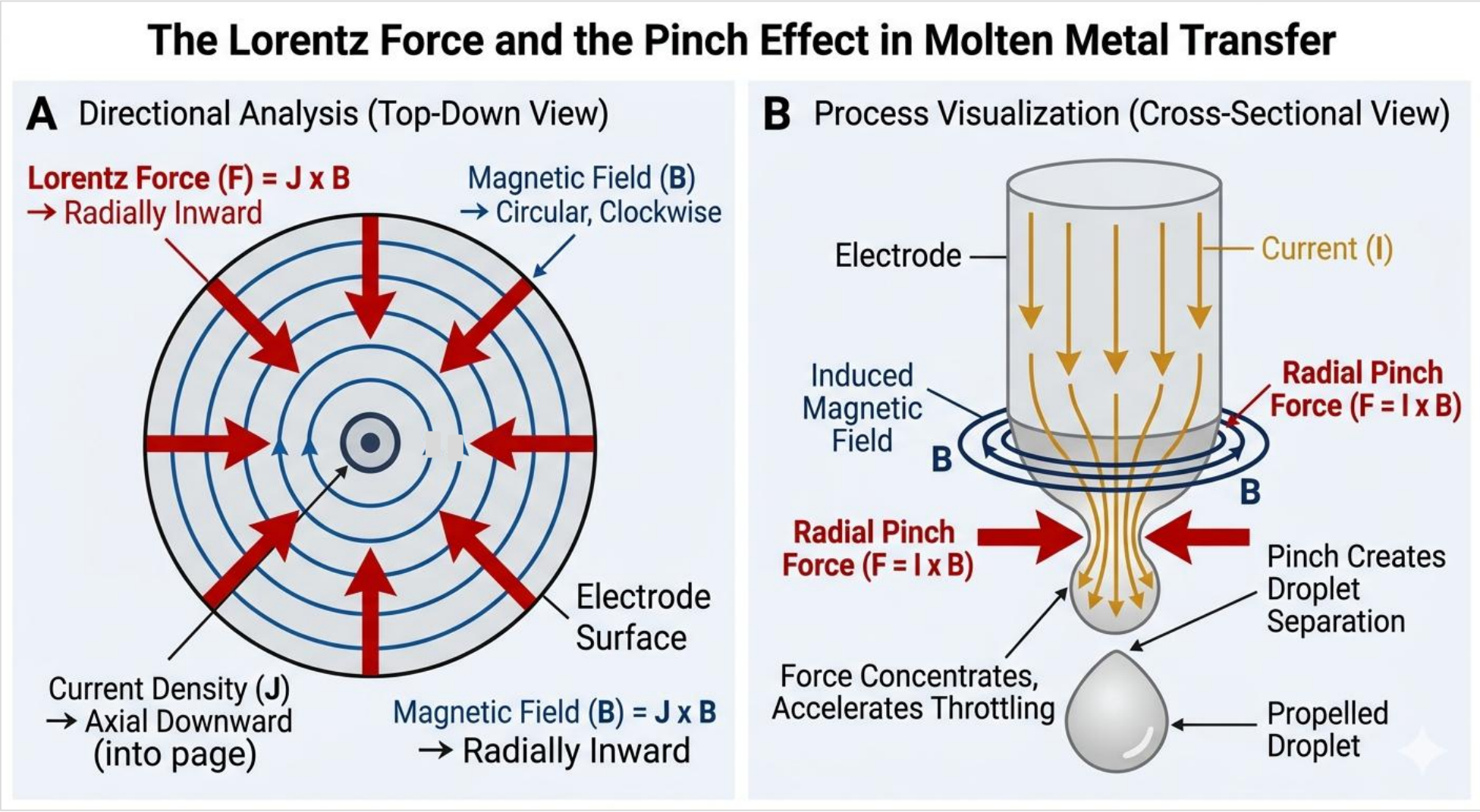
Wire Arc Additive Manufacturing (WAAM)

The Pinch Effect on Metal Droplet - DCEN



Wire Arc Additive Manufacturing (WAAM)

The Pinch Effect on Metal Droplet - DCEP



Wire Arc Additive Manufacturing (WAAM)

Short Circuit Transfer

Current Range : 100–130 A (Mild Steel) to 160 A (Stainless Steel).

Physics: At these lower currents, the Electromagnetic Pinch Force is too weak to overcome Surface Tension. The wire must physically touch the melt pool to transfer metal.

DED Application: Used for thin-walled structures where high heat would cause the part to collapse or warp

Wire Arc Additive Manufacturing (WAAM)

Globular Transfer

Current Range: ~130–200 A.

Physics: As current increases, the pinch force grows but remains insufficient to "neck" the wire quickly. Droplets grow large, often exceeding the wire diameter, until Gravity finally pulls them down.

DED Application: Not so preferable. May cause spatter

Wire Arc Additive Manufacturing (WAAM)

Spray Transfer

Current : ~200–240 A, typically for 1.2 mm wire.

Physics : At this point, the Electromagnetic Pinch Force becomes stronger than Surface Tension. It snaps the molten tip into hundreds of tiny droplets per second.

DED Application: High-productivity deposition for large, thick components

Wire Arc Additive Manufacturing (WAAM)

Pulsed Spray Transfer

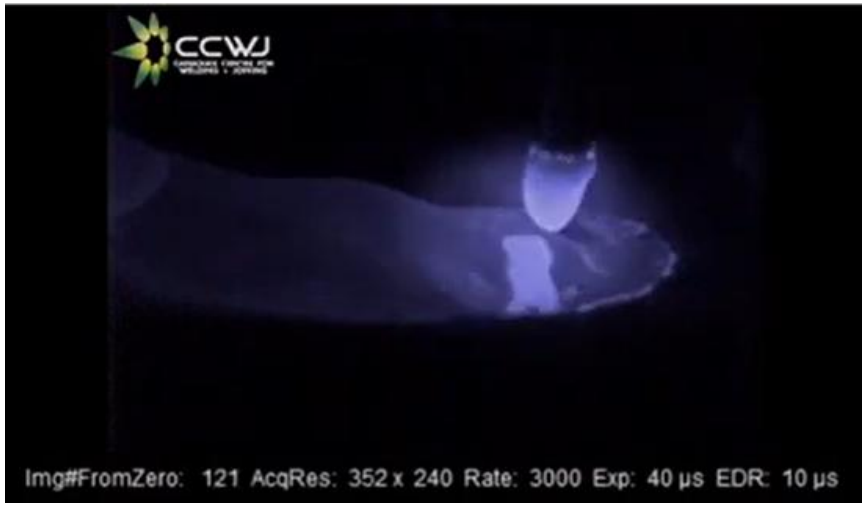
- ✓ Unlike standard spray transfer, which uses a constant high current, pulsed spray uses pulsed current
- ✓ When the current (power) spikes at peak level of pulse, the Electromagnetic Pinch Force becomes dominant. This "pinches" off molten droplet and propels it across the arc gap into the melt pool.
- ✓ As the power drops to the lower end of pulse, it is just enough to maintain the arc but too low to melt the wire or transfer the metal. This allows the melt pool to cool momentarily before the next pulse.

Peak Current : 300-600 A & Background Current (Low Level) : 30-150 A (typical)

Advantages of Pulsed Spray Transfer

- ✓ Reduced thermal distortion due to low average heat input
- ✓ Typically one droplet per pulse, making highly controlled droplet transfer & low spatter
- ✓ More consistent bead geometry
- ✓ Background current allows the melt pool to relax between pulses, and thus minimizes the turbulence
- ✓ Lower heat during background phase makes less penetration into substrate & reduced dilution
- ✓ Spray transfer in pulsed mode is cleaner than globular/short-circuit modes
- ✓ Since there is a cooling period in background current, easier to print vertical thin walls, avoids sagging in overhead builds, humping in horizontal prints etc.

Wire Arc Additive Manufacturing (WAAM)

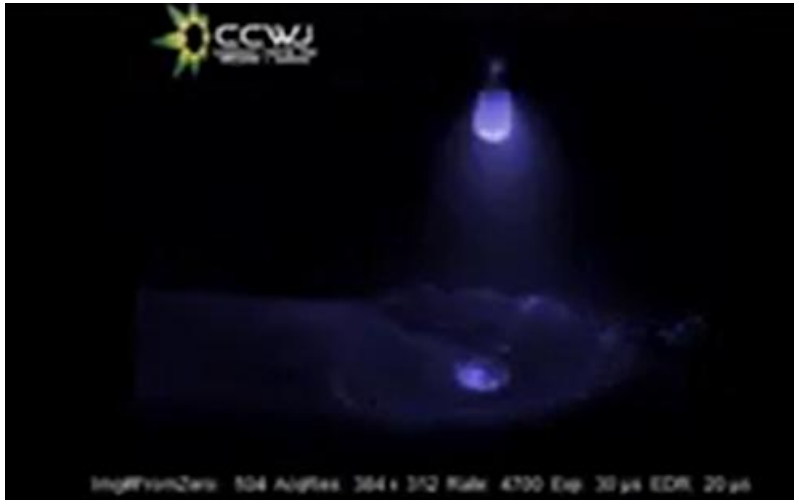


WAAM : MIG/ TIG System

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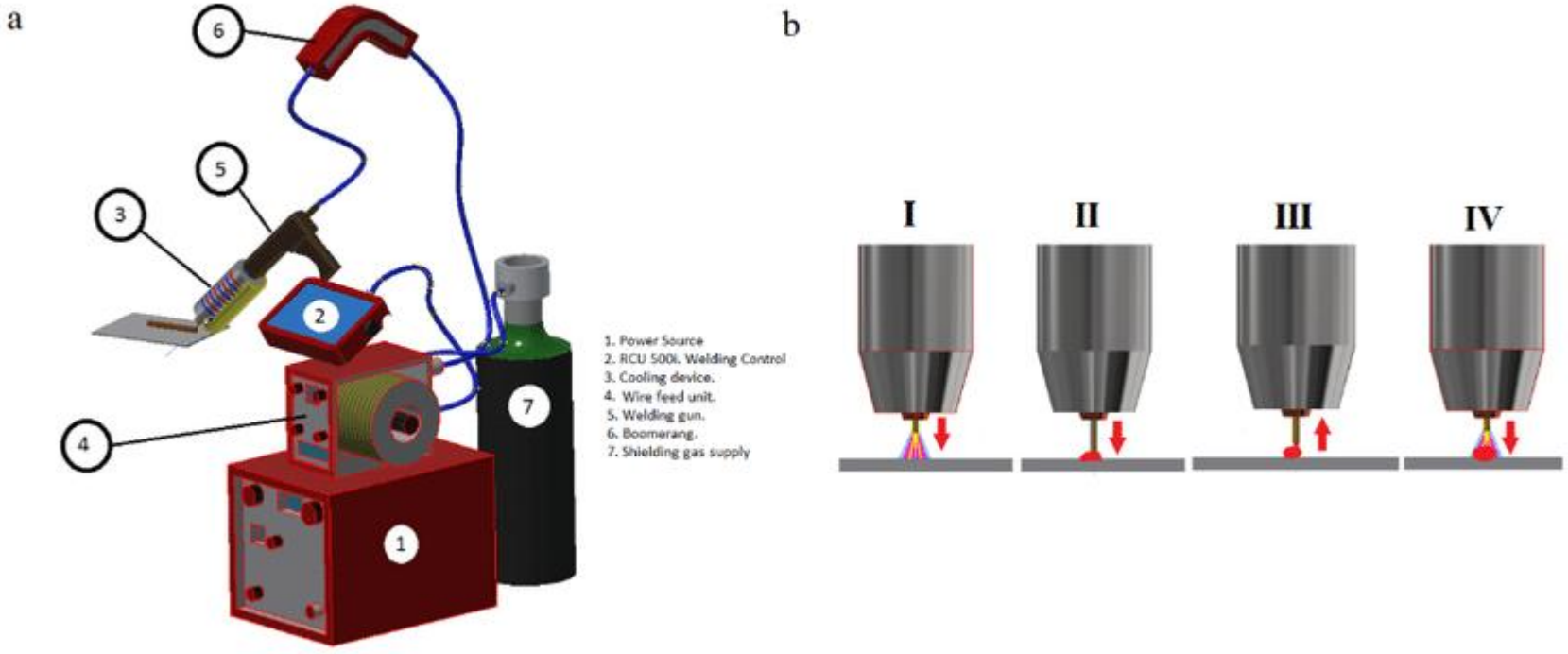






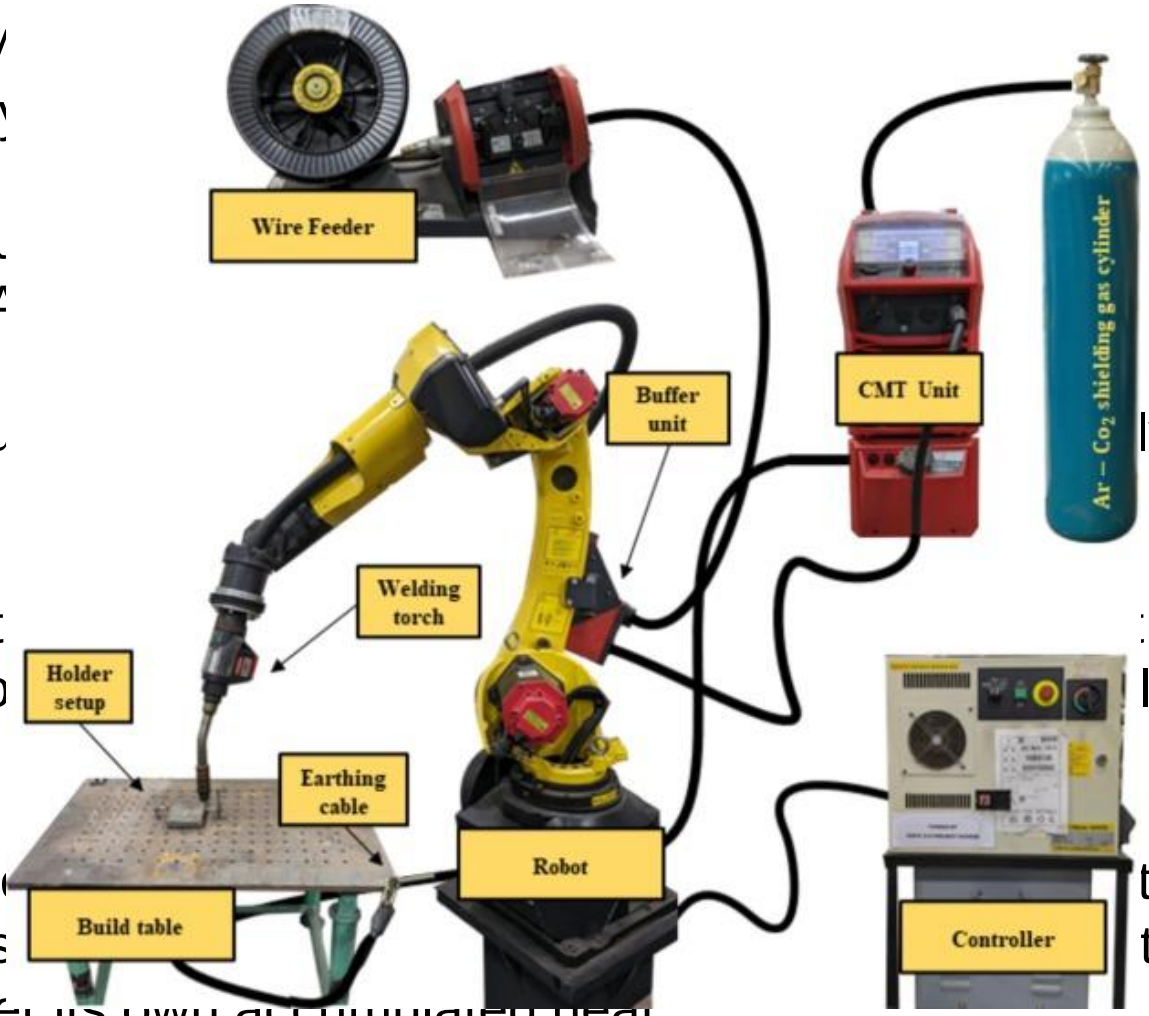
COLD METAL TRANSFER (CMT)

- ✓ Cold Metal Transfer (CMT) is a low-heat, modified WAAM process that alternates wire feeding and retraction to deposit material drop-by-drop.
- ✓ It acts by retracting the wire upon detecting a short circuit, allowing for controlled material deposition, reduced distortion, and superior joining of dissimilar materials (e.g., steel and aluminum) compared to traditional cladding

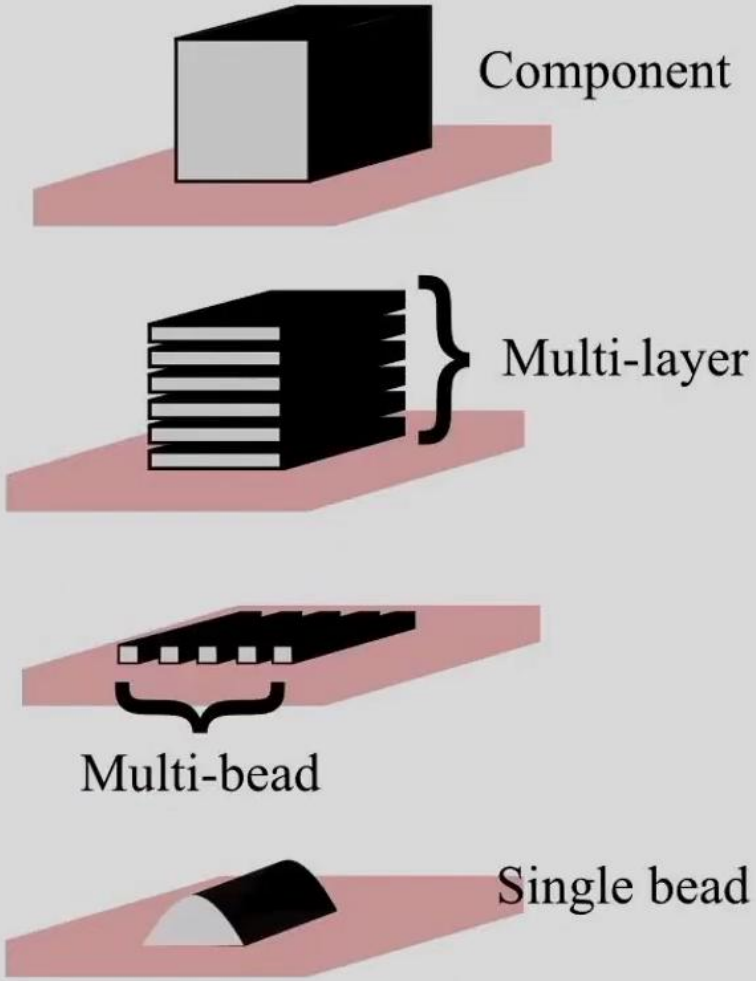


COLD METAL TRANSFER

- ✓ Unlike conventional GM to detach droplets, CMT integrates the wire's phy
- ✓ As soon as a short circu power source sends a signal to a high-speed A
- ✓ Instead of increasing cu ly reverses the wire feed direction (retracting the
- ✓ During this retraction, t is "peeled" into the weld pool by mechanical fo lown off by an electrical explosion.
- ✓ Because the detachme th 1/10th of the heat of conventional MIG. This tructures without the part melting or warping under its own accumulated heat.



WAAM: Major Process Variables

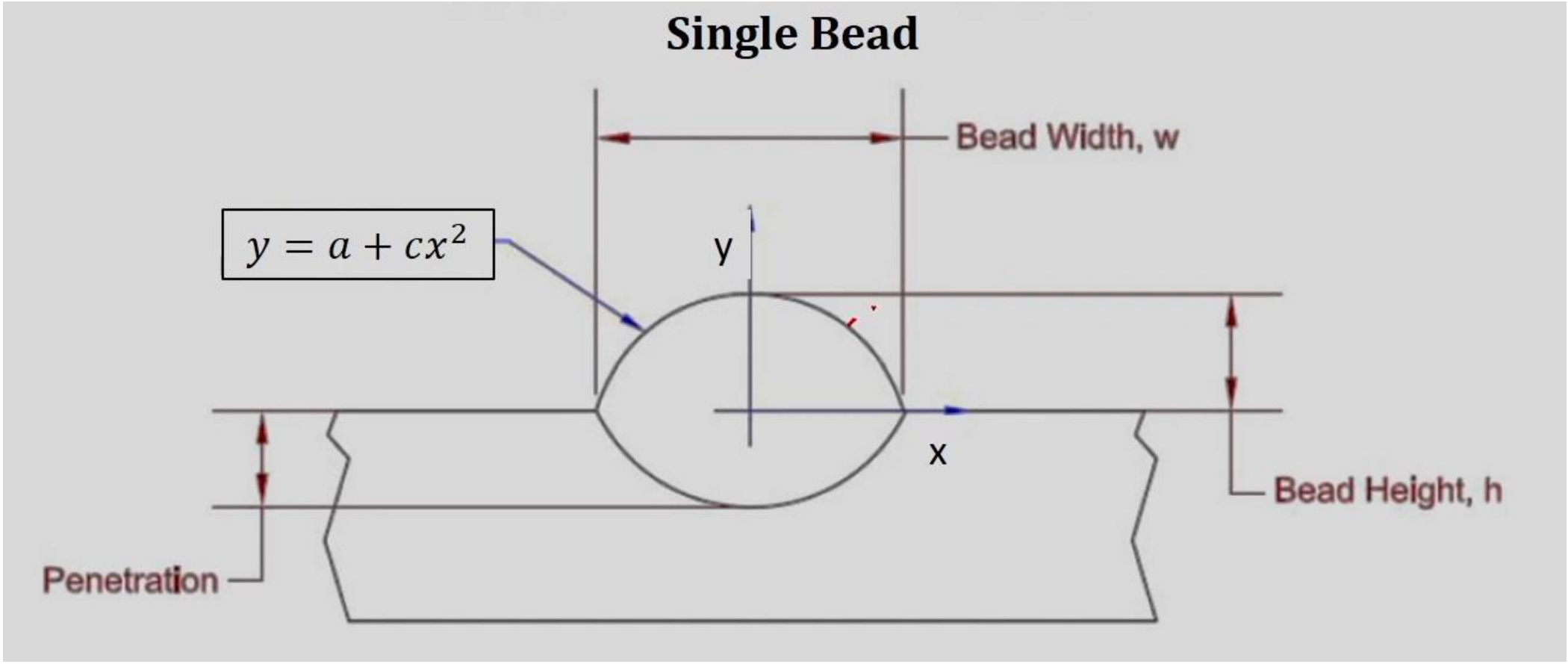


Power	**Arc voltage** **Starting, main, reduced and final current**
Filler	**Wire type (material composition, coating, coring etc.)** **Wire diameter** **Wire feed rate**
Gas	**Gas proportion, Gas pressure. Gas flow rate, Gas pre-flow period, Gas post-flow period**
Deposition	**Layer thickness** **Bead width** **Bead height or reinforcement height**
Torch	**Distance between nozzle and base plate** **Distance between electrode contact tip and base plate** **Torch speed**
Path	**Type of area-filling** **Step over increment**

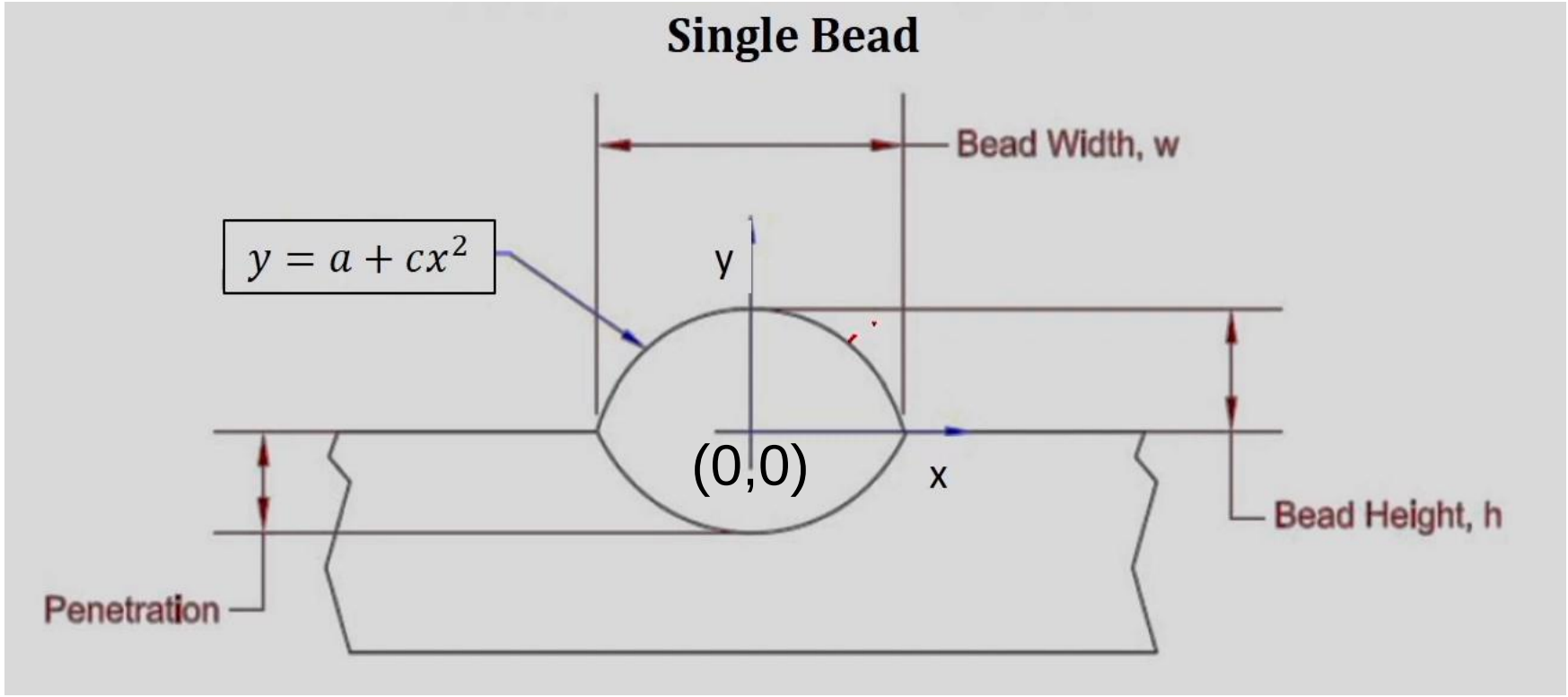
WAAM: Major Process Variables

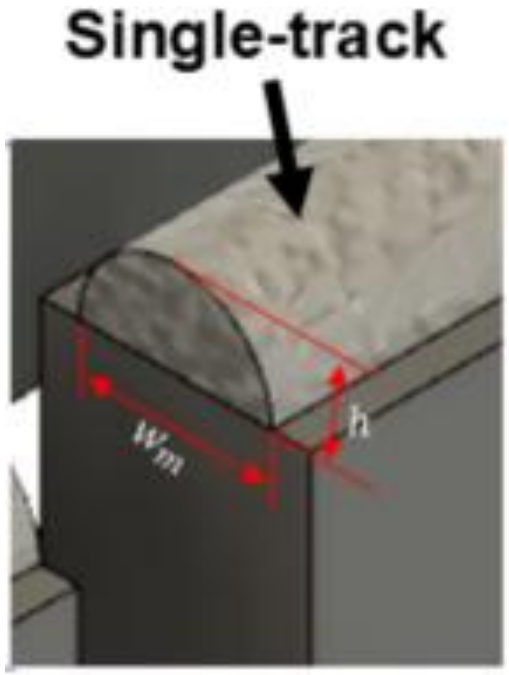
Single Bead Geometric Modelling

Assumption: Shape of the bead is parabola



WAAM: Single bead Modelling







WAAM: Single bead Modelling

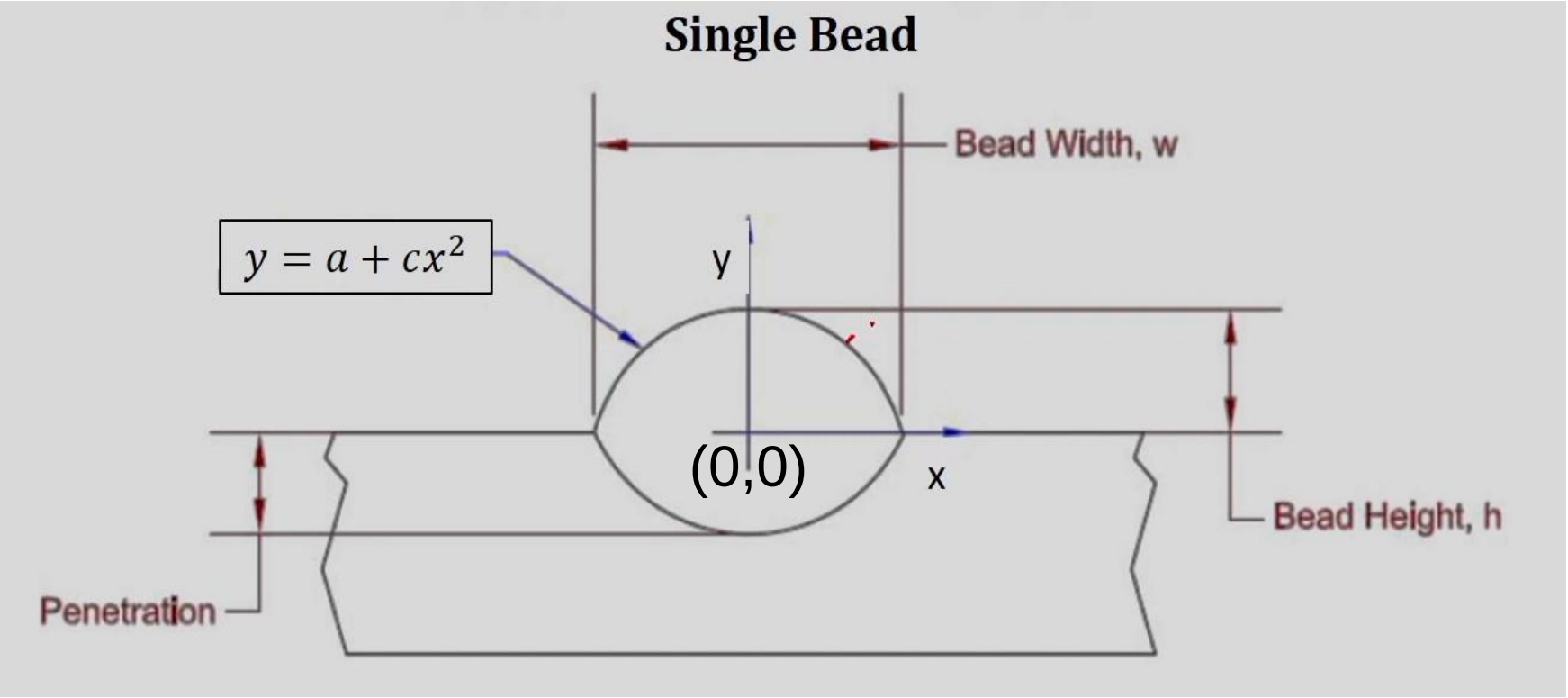
at x = 0, y = h

h - height of bead

at y=0 & x =0

$$0 = h + c\left(\frac{w}{2}\right)^2$$

$$c = -\frac{4h}{w^2}$$



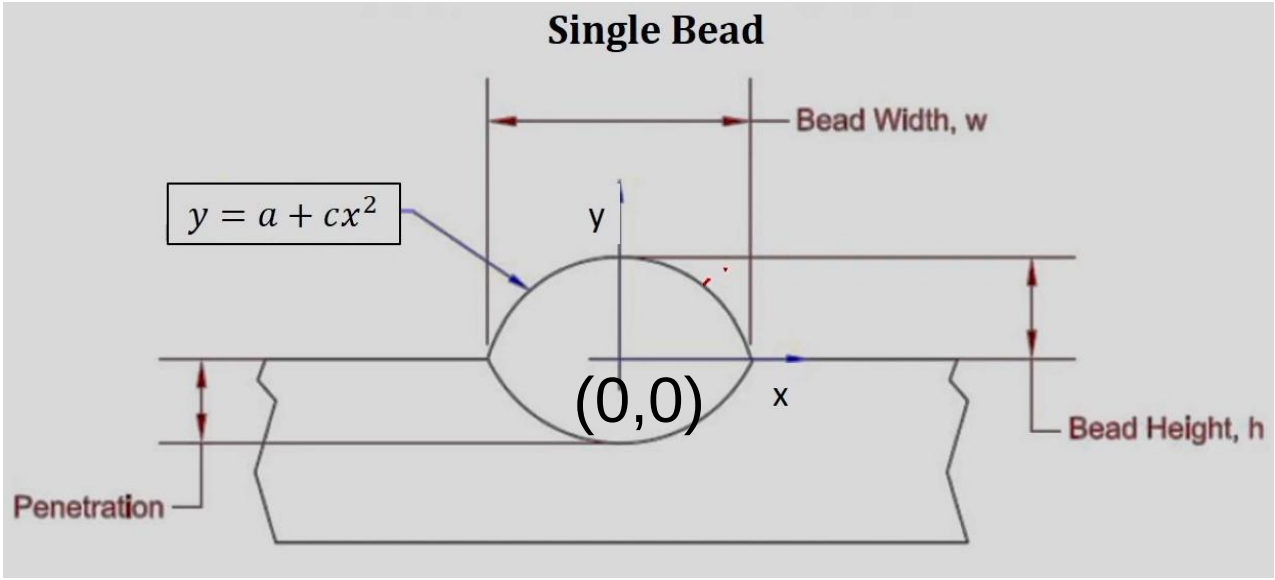
$$y = h - \left[\frac{4h}{w^2}x^2\right]$$

$$y = h\left[1 - \left[\frac{2x}{w}\right]^2\right]$$

WAAM: Single bead Modelling

Area of parabola from its geometry

$$A_{bead} = \frac{2}{3} h\, w$$



Volume flow rate Q

$$Q = A_{bead}\, V_{torch}$$

$$Q = \frac{2}{3} hw V_{torch}$$

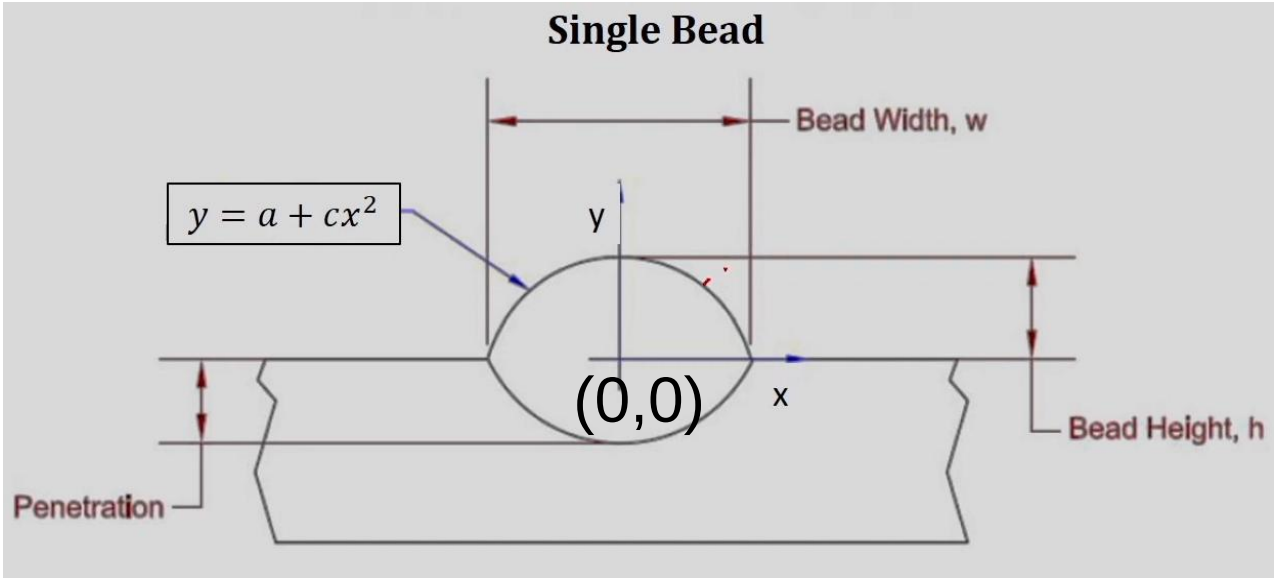
Volume flow rate also can be written as $Q = \frac{\pi d_{wire}^2}{4} V_{wire}$

Comparing both; $w = \frac{3\pi\, d_{wire}^2\, V_{wire}}{8\, h\, V_{torch}}$

WAAM: Single bead Modelling

Bead Profile, in terms of measurable height and selected process variables

$$y = h\left[1 - \left[\frac{16\,h\,V_{torch}\,x}{3\,\pi\,d_{wire}^{2}\,V_{wire}}\right]^{2}\right]$$



In many of the WAAM systems, wire feed is linearly proportional to current